CAREER POINT

JEE Main Exam 2016

(Paper & Solution)

Code – E Date : 03-04-2016

Part A – PHYSICS

Q.1 A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be:

(1) 92 ± 2 s (2) 92 ± 5.0 s (3) 92 ± 1.8 s (4) 92 ± 3 s

Ans. **[1]**

Sol. Time period of 100 oscillations

90 sec, 91 sec, 95 sec, 92 sec

Mean value of time = $\frac{90+91+95+92}{4}$

$t_m = 92$ sec

Absolute error

$|\Delta t_1| = |t_m - t_1| = 2$ sec

$|\Delta t_2| = |t_m - t_2| = 1$ sec

$|\Delta t_3| = |t_m - t_3| = 3$ sec

$|\Delta t_4| = |t_m - t_4| = 0$ sec

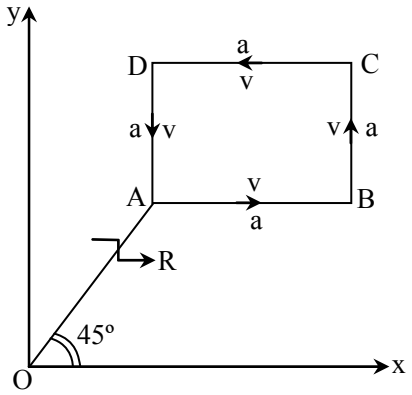
mean absolute error

$\Delta t_{mean} = \frac{2+1+3+0}{4}$

$\Delta t_{mean} = 1.5$ sec $\approx$ 2s

mean time (92 ± 2 sec)

Q.2 A particle of mass m is moving along side of a square of side 'a', with a uniform speed v in the x-y plane as shown in the figure:



Which of the following statements is **false** for the angular momentum $\vec{L}$ about the origin ?

(1) $\vec{L} = -\frac{mv}{\sqrt{2}} R\,\hat{k}$ when the particle is moving from A to B

(2) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} - a\right]\hat{k}$ when the particle is moving from C to D.

(3) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} + a\right]\hat{k}$ when the particle is moving from B to C.

(4) $\vec{L} = \frac{mv}{\sqrt{2}} R\,\hat{k}$ when the particle is moving from D to A.

Ans. **[2, 4]**

Sol. When particle move along AB

$\vec{L} = mvr_{\perp}(-\hat{k})$

$\vec{L} = mg\frac{R}{\sqrt{2}}(-\hat{k})$

when particle move alobng C to D

$\vec{L} = mvr_{\perp}(\hat{k})$

$\vec{L} = mv\left[\frac{R}{\sqrt{2}} + a\right](\hat{k})$

when particle move from B to C

$\vec{L} = mvr_{\perp}(\hat{k}) = mv\left(\frac{R}{\sqrt{2}} + a\right)(\hat{k})$

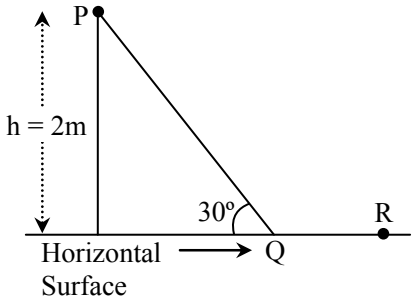
when particle move from D to A

$\vec{L} = mvr_{\perp}(-\hat{k}) = mv\frac{R}{\sqrt{2}}(-\hat{k})$

so (2) and (4)are false

Q.3 A point particle of mass m, moves along the uniformly rough track PQR as shown in the figure. The coefficient of friction, between the particle and the rough track equals μ. The particle is released, from rest, from the point P and it comes to rest at a point R. The energies, lost by the ball, over the parts, PQ and QR, of the track, are equal to each other, and no energy is lost when particle changes direction from PQ to QR.

The values of the coefficient of friction μ and the distance x(= QR), are, respectively close to:



(1) 0.2 and 6.5 m (2) 0.2 and 3.5 m (3) 0.29 and 3.5 m (4) 0.29 and 6.5 m

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Work Energy & power, Exercise # 4, Page No.41, Q.32]

Ans. **[3]**

Sol. From work-energy theorem

$W_g + W_{f_1} + W_{f_2} = \Delta K$

$mgh + 2W = 0 \quad \left[\because W_{f_1} = W_{f_2} = W\right]$

$W = -\frac{mgh}{2}$

and $W_{f_1} = -\frac{mgh}{2}$

$-\mu mg\cos30^\circ(4) = -\frac{mgh}{2}$

$\mu = 0.29$

and $W_{f_2} = +\frac{mgh}{2}$

$-\mu mgx = -\frac{mgh}{2}$

$x = \frac{h}{2\mu} = 3.5\ m$

Q.4 A person trying to lose weight by burning fat lifts a mass of 10 kg upto a height of 1m 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up ? Fat supplies 3.8×10^7 J of energy per kg which is converted to mechanical energy with a 20% efficiency rate. Take $g = 9.8\ ms^{-2}$:

(1) 2.45×10^{-3} kg (2) 6.45×10^{-3} kg (3) 9.89×10^{-3} kg (4) 12.89×10^{-3} kg

Ans. **[4]**

Sol. PE = 1000 × mgh

= 1000 × 10 × 9.8 × 1

= 98000 J

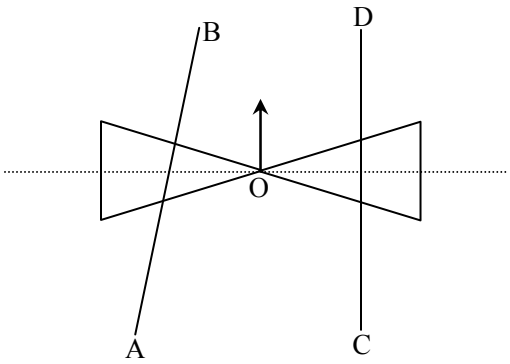
machenical energy per unit mass = $\frac{2}{100} \times 3.8 \times 10^7$

$= 7.6 \times 10^5$ J/kg

$\therefore$ Fat burn $= \frac{9800}{7.6 \times 10^5}$ kg

$= 12.89 \times 10^{-3}$ kg

Q.5 A roller is made by joining together two cones at their vertices O. It is kept on two rails AB and CD which are placed asymmetrically (see figure), with its axis perpendicular to CD and its centre O at the centre of line joining AB and CD (see figure). It is given a light push so that it starts rolling with its centre O moving parallel to CD in the direction shown. As it moves, the roller will tend to :



(1) turn left.

(2) turn right

(3) go straight

(4) turn left and right alternately.

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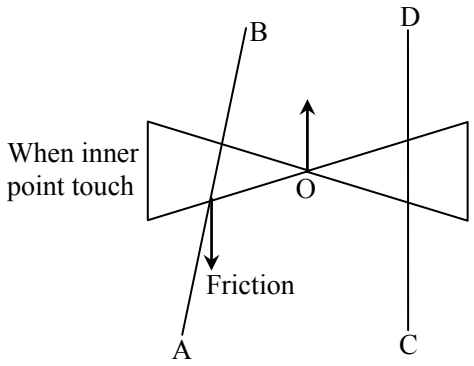
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Ans. **[1]**

Sol.



When this cone proceeds then on AB rail the inner point-will touch the rail whose velocity in forward direction which slip forward and friction acts in backward direction. Due to this friction the system turn left.

Q.6 A satellite is revolving in a circular orbit at a height 'h' from the earth's surface (radius of earth R; h << R). The minimum increases in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to : (Neglect the effect of atmosphere.)

(1) $\sqrt{2gR}$ (2) $\sqrt{gR}$ (3) $\sqrt{gR/2}$ (4) $\sqrt{gR}(\sqrt{2}-1)$

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter : Gravitation, Theory #12, Page No. 12, Q.g(iii) + d(iii)]
[JEE Advance, Chapter : Gravitation, Exercise # 5B, Page No. 28, Q.8]

Ans. **[4]**

Sol. Orbital velocity of a satellite revolving in a circular orbit at a height (h << R)

$v_0 = \sqrt{Rg}$

and the velocity required to escape is

$v_e = \sqrt{2Rg}$

So increase in velocity

$\Delta v = v_e - v_0 = (\sqrt{2}-1)\sqrt{Rg}$

Q.7 A pendulum clock loses 12 s a day if the temperature is 40ºC and gains 4s a day if the temperature is 20ºC. The temperature at which the clock will show correct time, and the co-efficient of linear expansion (α) of the metal of the pendulum shaft are respectively :

(1) 25ºC; $\alpha = 1.85 \times 10^{-5}$/ºC (2) 60ºC; $\alpha = 1.85 \times 10^{-4}$/ºC

(3) 30ºC; $\alpha = 1.85 \times 10^{-3}$/ºC (4) 55ºC; $\alpha = 1.85 \times 10^{-2}$/ºC

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter : Thermal Expansion, Exercise # 1, Page No.61, Q.10]
[JEE Advance, Chapter : Thermal Expansion, Exercise # 4, Page No.87, Q.15]

Ans. **[1]**

Sol. Let at temperature θ, clock gives correct time

$\Delta T = \left(\frac{1}{2}\alpha\Delta\theta\right)T, \qquad T = 1 \text{ day} = 86400s$

$12 = \frac{1}{2}\alpha\,(40-\theta)T \quad \ldots (i)$

$4 = \frac{1}{2}\alpha\,(\theta-20)\,T \quad \ldots(ii)$

$\frac{12}{4} = \frac{40-\theta}{\theta-20}$

$3\theta - 60 = 40 - \theta$

$\theta = 25^{\circ}C$

from (ii) $4 = \frac{1}{2}\alpha\,(25-20)\times 86400$

$\alpha = \frac{8}{5\times 86400} = 1.85\times 10^{-5}/^{\circ}C$

Q.8 An ideal gas undergoes a quasi static, reversible process in which its molar heat capacity C remains constant. If during this process the relation of pressure P and volume V is given by PV^n = constant, then n is given by (Here C_P and C_V are molar specific heat at constant pressure and constant volume, respectively):

(1) $n = \frac{C_P}{C_V}$ (2) $n = \frac{C-C_P}{C-C_V}$ (3) $n = \frac{C_P-C}{C-C_V}$ (4) $n = \frac{C-C_V}{C-C_P}$

Students may find similar question in CP exercise sheet :
[JEE Advance, Chapter : Thermodynamics, Exercise # 1, Page No. 25, Q.24]

Ans. **[2]**

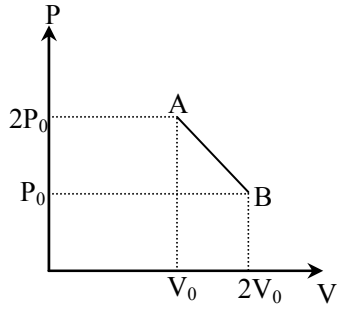
Sol. $C = C_V + \frac{PdV}{ndT} = \frac{R}{\gamma-1} + \frac{R}{1-n}$

$C = \frac{R}{\frac{C_P}{C_V}-1} + \frac{R}{1-n}$

on solving

$\boxed{n = \frac{C-C_P}{C-C_V}}$

Q.9 'n' moles of an ideal gas undergoes a process A → B as shown in the figure. The maximum temperature of the gas during the process will be:

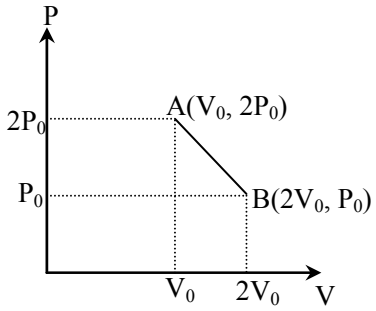


(1) $\frac{9P_0V_0}{4nR}$ (2) $\frac{3P_0V_0}{2nR}$ (3) $\frac{9P_0V_0}{2nR}$ (4) $\frac{9P_0V_0}{nR}$

Students may find same question in CP exercise sheet :
[*JEE Main, Chapter : KTG, Exercise # 4, Page No.28, Q.11*]
[*JEE Advance, Chapter : Thermodynamics, Exercise # 6, Page No. 49, Q.17*]

Ans. **[1]**

Sol.



$$P - 2P_0 = \frac{P_0 - 2P_0}{2V_0 - V_0}(V - V_0)$$

$$P = 2P_0 - \frac{P_0}{V_0}V + P_0$$

$$P = -\left(\frac{P_0}{V_0}\right)V + 3P_0$$

$$\frac{nRT}{V} = -\left(\frac{P_0}{V_0}\right)V + 3P_0$$

$$T = \frac{1}{nR}\left[-\frac{P_0}{V_0}V^2 + 3P_0V\right]$$

for T_{max}, $\frac{dT}{dV} = 0$

$$\frac{1}{nR}\left[-\frac{2P_0}{V_0}V + 3P_0\right] = 0$$

$$V = \frac{3}{2}V_0$$

$$\therefore \quad T_{max} = \frac{1}{nR}\left[-\frac{P_0}{V_0}\times\left(\frac{3}{2}V_0\right)^2 + 3P_0 \times \frac{3}{2}V_0\right]$$

$$= \frac{1}{nR}\left[-\frac{9}{4}P_0V_0 + \frac{9}{2}P_0V_0\right] = \frac{9P_0V_0}{4nR}$$

Q.10 A particle performs simple harmonic motion with amplitude A. Its speed is trebled at the instant that it is at a distance $\frac{2A}{3}$ from equilibrium position. The new amplitude of the motion is:

(1) $\frac{A}{3}\sqrt{41}$ (2) 3A (3) $A\sqrt{3}$ (4) $\frac{7A}{3}$

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : SHM, Exercise # 1, Page No.25, Q.17]

[JEE Advance, Chapter : SHM, Exercise # 5A, Page No. 34, Q.10]

Ans. **[4]**

Sol. Velocity at any position is given by

$$v = \omega\sqrt{A^2 - x^2}$$

at $x = \frac{2A}{3}$

$$v = \omega\sqrt{A^2 - \frac{4A^2}{9}} \Rightarrow v = \frac{\sqrt{5}}{3}A\omega$$

if speed at $x = \frac{2A}{3}$ is tripled i.e.

$v' = 3v = \sqrt{5}\,A\omega$

So new amplitude A′ is

$$v' = \omega\sqrt{A'^2 - \left(\frac{4A}{3}\right)^2}$$

$$\sqrt{5}\,A\omega = \omega\sqrt{A'^2 - \frac{4A^2}{9}}$$

$$A' = \frac{7A}{3}$$

Q.11 A uniform string of length 20m is suspended from a rigid support. A short wave pulse is introduced at its lowest end. It starts moving up the string. The time taken to reach the support is: (take $g = 10\ ms^{-2}$)

(1) $2\pi\sqrt{2}\,s$ (2) 2 s (3) $2\sqrt{2}\,s$ (4) $\sqrt{2}\,s$

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter :Wave Motion, Exercise # 2, Page No.29, Q.8]

[JEE Advance, Chapter : Transverse waves, Exercise # 4, Page No.36, Q.15]

Ans. **[3]**

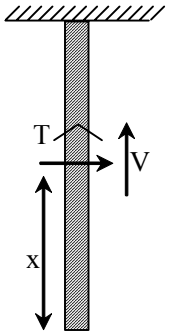
Sol. Let m → mass/length

T = (xm)g

$$V = \sqrt{\frac{T}{m}}$$

$$= \sqrt{\frac{(xm)g}{m}}$$

$$= \sqrt{gx}$$



$$a = \frac{VdV}{dx} = \sqrt{gx}\,\frac{d}{dx}\sqrt{gx}$$

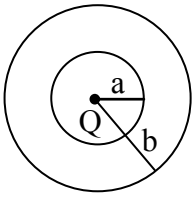
$$= \sqrt{gx} \times \sqrt{g} \times \frac{1}{2\sqrt{x}}$$

$$a = \frac{g}{2} = \text{constant}$$

$$\therefore \ell = \frac{1}{2}at^2$$

$$t = \sqrt{\frac{2\ell}{a}} = \sqrt{\frac{2\ell}{g/2}} = 2\sqrt{\frac{20}{10}} = 2\sqrt{2}s$$

Q.12 The region between two concentric spheres of radii 'a' and 'b', respectively (see figure), has volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q. The value of A such that the electric field in the region between the spheres will be constant, is

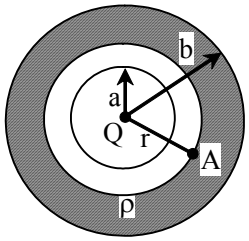


(1) $\frac{Q}{2\pi a^2}$ (2) $\frac{Q}{2\pi(b^2 - a^2)}$ (3) $\frac{2Q}{\pi(a^2 - b^2)}$ (4) $\frac{2Q}{\pi a^2}$

Ans. **[1]**

Sol. $\oint \vec{E} \cdot \vec{dS} = \frac{q_{in}}{\epsilon_0}$

$$E \cdot 4\pi r^2 = \frac{Q + \int_{r=a}^{r} \rho dV}{\epsilon_0}$$



$$E \cdot 4\pi r^2 = \frac{Q + \int_{a}^{r} \frac{A}{r} 4\pi r^2 dr}{\epsilon_0}$$

$$E = \frac{1}{4\pi \in_0 r^2}\left[Q + 4\pi A\left(\frac{r^2 - a^2}{2}\right)\right]$$

$$E = \frac{1}{4\pi \in_0 r^2}[Q + 2\pi A(r^2 - a^2)]$$

$$E = \frac{Q}{4\pi \in_0 r^2} + \frac{A(r^2 - a^2)}{2 \in_0 r^2} = \frac{Q}{4\pi \in_0 r^2} + \frac{A}{2 \in_0} - \frac{Aa^2}{2 \in_0 r^2}$$

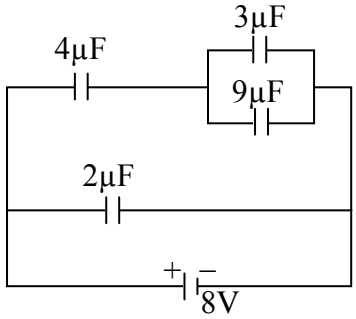
$\because$ E = constant

$\therefore$ E $\rightarrow$ Independent of r

$$\therefore \frac{Q}{4\pi \in_0 r^2} = \frac{Aa^2}{2 \in_0 r^2}$$

$$\boxed{A = \frac{Q}{2\pi a^2}}$$

Q.13 A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the 4 μF and 9 μF capacitors), at a point distant 30 m from it, would equal

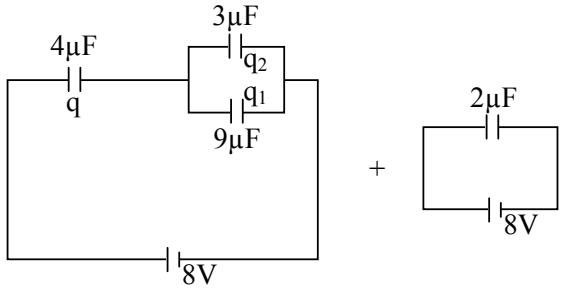


(1) 240 N/C (2) 360 N/C (3) 420 N/C (4) 480 N/C

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter : Capacitance, Exercise #2, Page No. 43, Q.18]
[JEE Advance, Chapter : Capacitance, Exercise # 2, Page No.24, Q.9]

Ans. **[3]**

Sol.



+

$$q = C_{eq}V = \left(\frac{4 \times 12}{4 + 12}\right) \times 8 = 24\ \mu C$$

$q_1 = \frac{9}{9+3} \times 24\ \mu C = 18\ \mu C$

$\therefore\ Q = q + q_1 = 24 + 18 = 42\ \mu C$

$E = \frac{KQ}{r^2} = \frac{9\times10^9 \times 42\times10^{-6}}{(30)^2} = 420\ N/C$

Q.14 The temperature dependence of resistances of Cu and undoped Si in the temperature range 300-400 K, is best described by

(1) Linear increase for Cu, linear increase for Si

(2) Linear increase for Cu, exponential increase for Si

(3) Linear increase for Cu, exponential decrease for Si

(4) Linear decrease for Cu, linear decrease for Si

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Current Electricty, Exercise #1, Page No.51, Q.15]

Ans. **[3]**

Sol. Cu is metal, its resistance increases linearly on increasing temperature, where as Si is semiconductor which resistance decreases exponentially on increasing temperature.

Q.15 Two identical wires A and B, each of length '*l*', carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side 'a'. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is

(1) $\frac{\pi^2}{8}$ (2) $\frac{\pi^2}{16\sqrt{2}}$ (3) $\frac{\pi^2}{16}$ (4) $\frac{\pi^2}{8\sqrt{2}}$

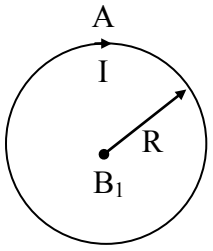
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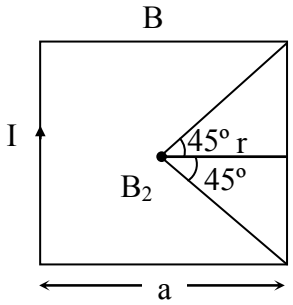
[JEE Main, Chapter :Magnetic effect of curernt, Exercise # 5B, Page No. 69, Q.61]

[JEE Advance, Chapter : Magnetic effect of current, Exercise # 1, Page No. 32, Q.9]

Ans. **[4]**

Sol.





$\ell = 2\pi R$ $\qquad$ $\ell = 4a$

$R = \frac{\ell}{2\pi}$ $\qquad$ $a = \frac{\ell}{4}$

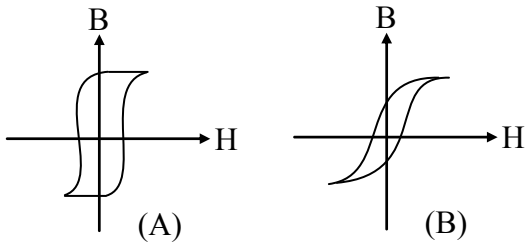
$$B_1 = \frac{\mu_0 I}{2R} = \frac{\mu_0 I}{2\times\frac{\ell}{2\pi}} = \frac{\mu_0 \pi I}{\ell}$$

$$B_2 = \frac{\mu_0 I}{4\pi r}[\sin\alpha + \sin\beta]\times 4 = \frac{\mu_0 I}{4\pi\frac{a}{2}}[\sin 45^\circ + \sin 45^\circ]\times 4$$

$$= 2\sqrt{2}\frac{\mu_0 I}{\pi a} = 2\sqrt{2}\frac{\mu_0 I}{\frac{\pi\ell}{4}} = 8\sqrt{2}\frac{\mu_0 I}{\pi\ell}$$

$$\frac{B_1}{B_2} = \frac{\frac{\mu_0\pi I}{\ell}}{8\sqrt{2}\frac{\mu_0 I}{\pi\ell}} = \frac{\pi^2}{8\sqrt{2}}$$

Q.16 Hysteresis loops for two magnetic materials A and B are given below :



These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use

(1) A for electric generators and transformers

(2) A for electromagnets and B for electric generators

(3) A for transformers and B for electric generators

(4) B for electromagnets and transformers

Ans. **[4]**

Sol. fig. A $\rightarrow$ Hard magnetic material

fig. B $\rightarrow$ soft magnetic material

Transformer core and electromagnets are made by soft magnetic materials, which are easy to magnetise and demagnetise where as for electric generators we use hard magnetic materials.

Q.17 An arc lamp requires a direct current of 10 A at 80 V to function. If it is connected to a 220 V (rms), 50 Hz AC supply, the series inductor needed for it to work is close to

(1) 80 H (2) 0.08 H (3) 0.044 H (4) 0.065 H

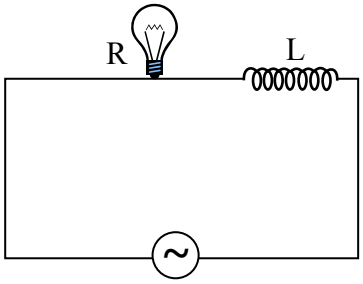
Students may find similar question in CP exercise sheet :

[JEE Main, Chapter :Alternating current, Exercise # 2, Page No. 37, Q.41]

[JEE Advance, Chapter : Alternating current, Exercise # 3, Page No. 36, Q.17]

Ans. **[4]**

Sol.



220V-50 Hz

Resistance of lamp $R = \frac{V}{i} = \frac{80}{10} = 8\ \Omega$

$V_R = 80$

$iR = 80$

$\left(\frac{220}{\sqrt{R^2 + X_L^2}}\right)R = 80$

$\sqrt{R^2 + X_L^2} = \frac{11}{4}R$

$R^2 + X_L^2 = \frac{121}{16}R^2$

$X_L^2 = \frac{105}{16}R^2 = \frac{105}{16} \times 8^2$

$X_L^2 = 420$

$X_L = 20.5$

$2\pi fL = 20.5$

$2 \times 3.14 \times 50\ L = 20.5$

$L = 0.065$ H

Q.18 Arrange the following electromagnetic radiations per quantum in the order of increasing energy :

A : Blue light B : Yellow light

C : X-ray D : Radiowave

(1) D, B, A, C (2) A, B, D, C (3) C, A, B, D (4) B, A, D, C

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Electromagnetic Wave, Exercise # 3, Page No. 101, Q.13]

Ans. **[1]**

Sol. D, B, A, C

Q.19 An observer looks at a distant tree of height 10 m with a telescope of magnifying power of 20. To the observer the tree appears :

(1) 10 times taller (2) 10 times nearer

(3) 20 times taller (4) 20 times nearer

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Electromagnetic wave, Exercise # 3, Page No.90, Q.13]

Ans. **[4]**

Sol. Telescope provides angular magnification i.e. angle formed by image is more than that of object and it forms image closer to the instrument so angle formed by image is increased.

Hence Ans.(4)

i.e., 20 times nearer

Q.20 The box of a pin hole camera, of length L, has a hole of radius *a*. It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say b_{min}) when :

(1) $a = \frac{\lambda^2}{L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$

(2) $a = \sqrt{\lambda L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$

(3) $a = \sqrt{\lambda L}$ and $b_{min} = \sqrt{4\lambda L}$

(4) $a = \frac{\lambda^2}{L}$ and $b_{min} = \sqrt{4\lambda L}$

Ans. **[3]**

Sol. Width of CM $= \frac{2D\lambda}{a}$

$= \frac{2L\lambda}{a} = \frac{2L\lambda}{\sqrt{\lambda L}}$

$= \sqrt{4\lambda L}$

Q.21 Radiation of wavelength λ, is incident on a photocell. The fastest emitted electron has speed v. If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be :

(1) $> v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

(2) $< v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

(3) $= v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

(4) $= v\left(\frac{3}{4}\right)^{\frac{1}{2}}$

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Photoelectric effect, Exercise # 5B, Page No.36, Q.16*]

[*JEE Advance, Chapter : Photoelectric effect, Exercise # 3, Page No.24, Q.2*]

Ans. **[1]**

Sol. $\frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi$ .... (i)

$$\frac{1}{2}m(v')^2 = \frac{hc}{\frac{3\lambda}{4}} - \phi$$

$$\frac{1}{2}m(v')^2 = \frac{4}{3}\frac{hc}{\lambda} - \phi$$

$$\frac{1}{2}m(v')^2 = \frac{4}{3}\left(\frac{1}{2}mv^2 + \phi\right) - \phi$$

$$\frac{1}{2}m(v')^2 = \frac{4}{3}\left(\frac{1}{2}mv^2\right) + \frac{\phi}{3}$$

$$\frac{1}{2}m(v')^2 > \frac{4}{3}\left(\frac{1}{2}mv^2\right)$$

$$\therefore v' > \sqrt{\frac{4}{3}}v$$

Q.22 Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed numbers of A and B nuclei will be :

(1) 1 : 16 (2) 4 : 1 (3) 1 : 4 (4) 5 : 4

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Nuclear physics & Radioactivity, Exercise # 1, Page No. 29, Q.22]

Ans. **[4]**

Sol.

$A\left(T_{\frac{1}{2}} = 20\,\text{min}\right)$ $B\left(T_{\frac{1}{2}} = 40\,\text{min.}\right)$

N_0 N_0

$\downarrow$ $\downarrow$

$\frac{N_0}{2}$ $\frac{N_0}{2}$

$\downarrow$ $\downarrow$

$\frac{N_0}{4}$ $\frac{N_0}{4}$

$\downarrow$ No of decayed nuclei

$\frac{N_0}{8}$ $= N_0 - \frac{N_0}{4} = \frac{3N_0}{3}$

$\downarrow$

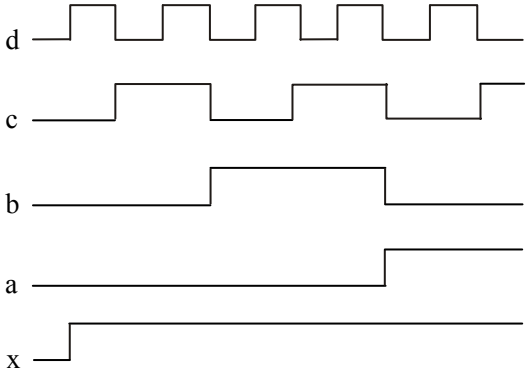
$\frac{N_0}{16}$

No of decayed nuclei

$= N_0 - \frac{N_0}{16} = \frac{15N_0}{16}$

Ratio of decayed Nuclei $= \frac{15N_0}{16} \Big/ \frac{3N_0}{4} = \frac{5}{4}$

Q.23 If a, b, c, d are inputs to a gate and x is its output, then, as per the following time graph, the gate is :



(1) NOT (2) AND (3) OR (4) NAND

Ans. **[3]**

Sol. If one of the inputs is high then output will be high in OR gate.

Q.24 Choose the correct statement :

(1) In amplitude modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal

(2) In amplitude modulation the frequency of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal

(3) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal

(4) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the frequency of the audio signal

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Communication, Exercise # 1, Page No.101, Q.7*]

Ans. **[1]**

Sol. (1) is correct but language of this option is not upto the mark.

Q.25 A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides the main scale line ?

(1) 0.75 mm (2) 0.80 mm (3) 0.70 mm (4) 0.50 mm

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Practical physics, Page No.26, Q.19*]

[*JEE Advance, Chapter : Practical physics, Exercise # 3, Page No.72, Q.5*]

Ans. **[2]**

Sol. $LC = \dfrac{\text{pitch}}{\text{no. of div. on circular scale}}$

$= \dfrac{0.5\,\text{mm}}{50} = 0.01$ mm

Zero error e = N × LC = 5 × 0.01

= 0.05 mm

True reading = observed reading + e

= LSR + CSR + e

= 0.5 mm + 25 × 0.01 + 0.05

= 0.75 + 0.05 = 0.80 mm

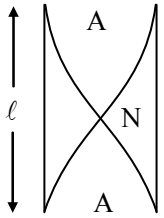
Q.26 A pipe open at both ends has a fundamental frequency f in air. The pipe is dipped vertically in water so that half of it is in water. The fundamental frequency of the air column is now :

(1) $\dfrac{f}{2}$ (2) $\dfrac{3f}{4}$ (3) $2f$ (4) f

Students may find same question in CP exercise sheet :
[JEE Main, Chapter :Sound wave, Exercise # 1, Page No.43, Q.26]
[JEE Advance, Chapter : Sound wave, Exercise # 1, Page No. 83, Q.14]

Ans. **[4]**

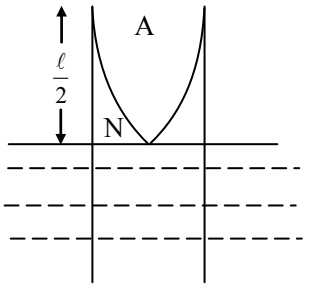
Sol.



$\ell = \dfrac{\lambda}{4} + \dfrac{\lambda}{4} = \dfrac{\lambda}{2}$

$\lambda = 2\ell$

$f = \dfrac{v}{\lambda} = \dfrac{v}{2\ell}$



$\dfrac{\ell}{2} = \dfrac{\lambda}{4} \Rightarrow \lambda = 2\ell$

$\therefore f' = \dfrac{v}{\lambda} = \dfrac{v}{2\ell} = f$

Q.27 A galvanometer having a coil resistance of 100 Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10 A, is :

(1) 0.01 Ω (2) 2 Ω (3) 0.1 Ω (4) 3 Ω

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter : Magnetic effect of current, Exercise # 1, Page No.45, Q.76]
[JEE Advance, Chapter : Current Electricity, Exercise # 2, Page No. 40, Q.28]

Ans. **[1]**

Sol. $G = 100\ \Omega$

$i_g = 1\ mA = 10^{-3}\ A$

$i = 10\ A$

$s = ?$

$$s = \frac{i_g \times G}{i - i_g}$$

$$= \frac{10^{-3} \times 100}{10 - 10^{-3}}$$

$$= \frac{10^{-3} \times 100}{10}$$

$$= 10^{-2}\ \Omega$$

Q.28 In an experiment for determination of refractive index of glass of a prism by i – δ, plot, it was found that a ray incident at angle 35º, suffers a deviation of 40º and that it emerges at angle 79º. In that case which of the following is closest to the maximum possible value of the refractive index ?

(1) 1.5 (2) 1.6 (3) 1.7 (4) 1.8

Ans. **[1]**

Sol. $i = 35^\circ$

$\delta = 40^\circ$

$e = 79^\circ$

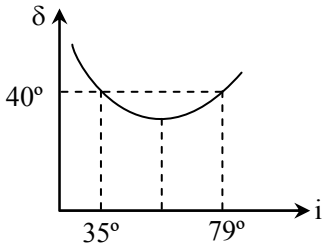
$A = ?$

$\delta = i + e - A$

$40^\circ = 35^\circ + 79^\circ - A$

$A = 35^\circ + 79^\circ - 40^\circ$

$A = 74^\circ$



$$\mu = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\frac{A}{2}}$$

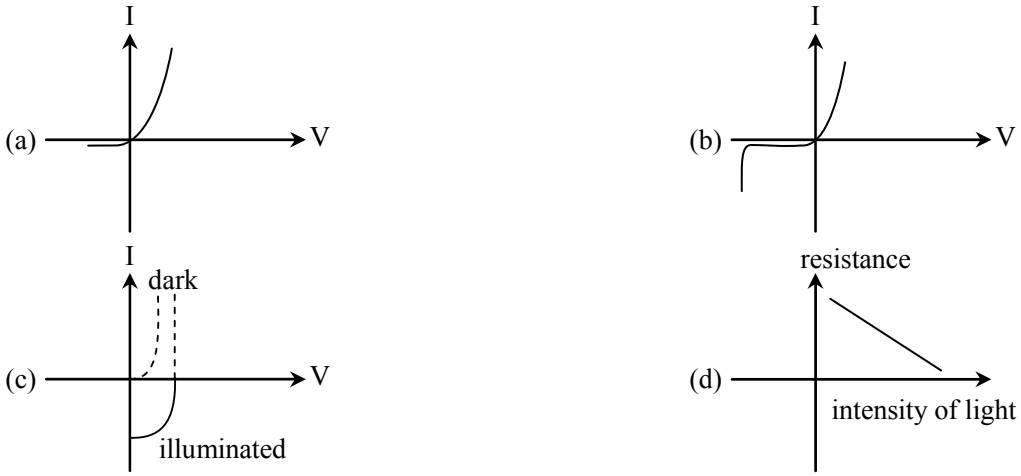
$$\mu = \frac{\sin\left(\frac{74^\circ+\delta_m}{2}\right)}{\sin 37^\circ}$$

$$\mu = \frac{5}{3}\sin\left(37^\circ+\frac{\delta_m}{2}\right)$$

$$\mu < \frac{5}{3}\sin 57^\circ < \frac{5}{3}\sin 60^\circ$$

$$\mu < \frac{5}{2\sqrt{3}} \Rightarrow \mu < 1.44$$

Q.29 Identify the semiconductor devices whose characteristics are given below, in the order (a), (b), (c), (d) :



(1) Simple diode, Zener diode, Solar cell, Light dependent resistance
(2) Zener diode, Simple diode, Light dependent resistance, Solar cell
(3) Solar cell, Light dependent resistance, Zener diode, Simple diode
(4) Zener diode, Solar cell, Simple diode, Light dependent resistance

Ans. **[1]**

Sol. By theory

Q.30 For a common emitter configuration, if α and β have their usual meanings, the incorrect relationship between α and β is :

(1) $\frac{1}{\alpha} = \frac{1}{\beta} + 1$ (2) $\alpha = \frac{\beta}{1-\beta}$

(3) $\alpha = \frac{\beta}{1+\beta}$ (4) $\alpha = \frac{\beta^2}{1+\beta^2}$

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter : Semiconductor Electronics, Theory: point no. 17, Page No. 22

Ans. **[2, 4]**

Sol. $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

or $\alpha = \frac{\beta}{1+\beta}$

Part B – CHEMISTRY

Q.31 At 300 K and 1 atm, 15 mL of a gaseous hydrocarbon requires 375 mL air containing 20% O_2 by volume for combustion. After combustion the gaseous occupy 330 mL. Assuming that the water formed is in liquid form and the volumes were measured at the same temperature and pressure, the formula of the hydrocarbon is :

(1) C_3H_6 (2) C_3H_8 (3) C_4H_8 (4) C_4H_{10}

Ans. **[Bonus]**

Sol. $$C_xH_y(g) + \left(x + \frac{y}{4}\right)O_2(g) \longrightarrow x\,CO_2(g) + \frac{y}{2}H_2O(\ell)$$

Initially	15 ml	75 ml	0	–
After reaction	0	0	15x	–

As per problem :

$$\frac{15}{1} = \frac{75}{\left(x + \frac{y}{4}\right)}$$

So, $x + \frac{y}{4} = 5$ ….(i)

and $15x = 330 - 300$

$x = 2$ ….(ii)

from eq. (i) & (ii)

$y = 12$

In this problem it seems to be there is a printing error if volume of gaseous products after combustion is 345 ml instead of 330 ml.

$15x = 345 - 300$

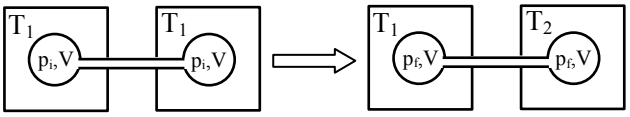
$x = 3$

& $x + \frac{y}{4} = 5$

$\therefore\ y = 8$

C_3H_8 Ans.

Q.32 Two closed bulbs of equal volume (V) containing an ideal gas initially at pressure p_i and temperature T_1 are connected through a narrow tube of negligible volume as shown in the figure below. The temperature of one of the bulbs is then raised to T_2. The final pressure p_f is :



(1) $p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$ (2) $2p_i\left(\frac{T_1}{T_1+T_2}\right)$ (3) $2p_i\left(\frac{T_2}{T_1+T_2}\right)$ (4) $2p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$

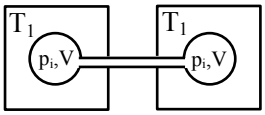
Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Gaseous State, Solved Example No. 32]

[JEE Advance, Chapter : Gaseous State, Exercise. # 2, Q.39]

Ans. **[3]**

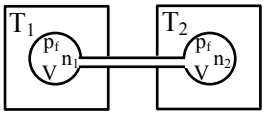
Sol. Initial condition



So, initial mole in both the bulb is same, say (n)

$$\text{So, } n = \frac{p_i V}{R\,T_i}$$

Final condition



So, $n_1 + n_2 = 2n$ [Total mole must be same]

$$\frac{P_f V}{RT_1} + \frac{P_f V}{RT_2} = 2\left(\frac{P_i V}{RT_1}\right)$$

$$P_f\left[\frac{1}{T_1} + \frac{1}{T_2}\right] = \frac{2P_i}{T_1}$$

$$P_f = 2P_i\left\{\frac{T_2}{T_1 + T_2}\right\}$$

Q.33 A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V esu. If e and m are charge and mass of an electron, respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by :

(1) meV (2) 2meV (3) $\sqrt{meV}$ (4) $\sqrt{2meV}$

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Atomic Structure, Key concepts, Topic :-Wave mechanical model of atoms*]

Students may find same question in CP exercise sheet :

[*JEE Advance, Chapter : Atomic structure, Same Q., Page No.3*]

Ans. **[4]**

Sol. $\lambda = \frac{h}{p}$ ….(1)

$p = mv$ ….(2)

$K.E. = \frac{1}{2}mv^2$

$v = \sqrt{\frac{2KE}{m}}$ …..(3)

from eq. (1), (2) & (3)

$$\lambda = \frac{h}{m\sqrt{\frac{2KE}{m}}}$$

$$\lambda = \frac{h}{\sqrt{2mKE}} \quad \ldots.(4)$$

If electron is accelerated by potential difference of V then its K.E. is given as follows

K.E. = eV ….(5)

from eq. (4) and (5)

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\frac{h}{\lambda} = \sqrt{2meV}$$

Q.34 The species in which the N atom is in a state of sp hybridization is :

(1) NO_2^+ (2) NO_2^- (3) NO_3^- (4) NO_2

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Chemical bonding, Exercise. # 4, Q.26]

[JEE Advance, Chapter : Chemical bonding, Exercise. # 2, Q.30]

Ans. **[1]**

Sol. In NO_2^+, 'N' has zero lone pair and two bond pairs (σ-bond). Hence, hybridization is sp.

Q.35 The heats of combustion of carbon and carbon monoxide are –393.5 and –283.5 kJ mol^{-1}, respectively. The heat of formation (in kJ) of carbon monoxide per mole is :

(1) 110.5 (2) 676.5 (3) –676.5 (4) –110.5

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Chemical energetics, Exercise. # 4, Q.6]

[JEE Advance, Chapter : Chemical energetics, Exercise. # 2, Q.12]

Ans. **[4]**

Sol. Given

$C(s) + O_2(g) \longrightarrow CO_2(g)$, $\Delta H_1 = -393.5$ kJ/mol

$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$, $\Delta H_2 = -283.5$ kJ/mol

Object reaction,

$C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g) \quad \Delta H = ?$

eq (i) – eq (ii)

$\Delta H = \Delta H_1 - \Delta H_2$

$= -393.5 + 283.5$

$= -110$ KJ/mol.

Q.36 18 g glucose ($C_6H_{12}O_6$) is added to 178.2 g water. The vapour pressure of water (in torr) for this aqueous solution is:

(1) 7.6 (2) 76.0 (3) 752.4 (4) 759.0

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Solution & Colligative properties, Exercise. # 4, Q.16*]

[*JEE Advance, Chapter : Solution & Colligative properties, Exercise. # 5A, Q.5*]

Ans. **[3]**

Sol. We assume the solution is prepared at boiling point temperature of water

$\therefore$ P° = 760 mm Hg

$$\frac{P° - P_s}{P°} = \frac{n}{n+N}$$

$$\frac{760 - P_s}{760} = \frac{0.1}{9.9+0.1}$$

$$1 - \frac{P_s}{760} = \frac{0.1}{10}$$

$$1 - 0.01 = \frac{P_s}{760}$$

P_s = 752.4 mm Hg or torr.

Q.37 The equilibrium constant at 298 K for a reaction A + B $\rightleftharpoons$ C + D is 100. If the initial concentration of all the four species were 1 M each, then equilibrium concentration of D (in mol L^{-1}) will be :

(1) 0.182 (2) 0.818 (3) 1.818 (4) 1.182

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Chemical equilibrium, Exercise. # 5B, Q.5*]

[*JEE Advance, Chapter : Chemical equilibrium, Exercise. # 2, Q.2*]

Ans. **[3]**

Sol.

	A	+	B	$\rightleftharpoons$	C	+	D
t = 0	1		1		1		1
eqm:	1–x		1–x		1+x		1+x

At the given state Q = 1; therefore Q < K

$\therefore$ reaction will move in forward direction.

So, $100 = \frac{(1+x)(1+x)}{(1-x)(1-x)}$

$$10 = \frac{1+x}{1-x}$$

$10 - 10x = 1 + x$

$11x = 9$

$$x = \frac{9}{11}$$

So, $[D] = 1 + x = 1 + \frac{9}{11}$

$$= \frac{20}{11} = 1.818 \text{ M}$$

Q.38 Galvanization is applying a coating of :

(1) Pb (2) Cr (3) Cu (4) Zn

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Electro chemistry, Key concept, Topic :- Prevention from corrosion*]

Ans. **[4]**

Sol. In Galvanization Zn coating is applied an iron.

$E^0_{OX\ Zn} > E^0_{OX\ Fe}$

Q.39 Decomposition of H_2O_2 follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125 M in one such decomposition. When the concentration of H_2O_2 reaches 0.05 M, the rate of formation of O_2 will be :

(1) 6.93×10^{-2} mol min^{-1} (2) 6.93×10^{-4} mol min^{-1}

(3) 2.66 L min^{-1} at STP (4) 1.34×10^{-2} mol min^{-1}

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Chemical kinetics, Exercise. # 4, Q.22*]

[*JEE Advance, Chapter : Chemical kinetics, Exercise. # 2, Q.43*]

Ans. **[2]**

Sol. Here solution is given assuming unit of rate is the option to be mol L^{-1} min^{-1}.

For 1st order reaction,

$$K = \frac{1}{50} \ln \frac{0.5}{0.125}$$

$$K = \frac{\ln 4}{50} min^{-1}$$

Now, $R = K [H_2O_2]^1$

$$R = \left(\frac{\ln 4}{50}\right) \times 0.05 \text{ M } min^{-1}$$

$R = 2 \times 0.693 \times 10^{-3}$ mol lit^{-1} min^{-1}

And, $H_2O_2 \longrightarrow H_2O + \frac{1}{2} O_2$

So rate of reaction, $R = \frac{R_{O_2} (\text{Rate of formation of } O_2)}{\left(\frac{1}{2}\right)}$

Or $R_{O_2} = \frac{1}{2} R = \frac{2 \times 0.693 \times 10^{-3}}{2}$ mol Lit^{-1} min^{-1}

$R_{O_2} = 6.93 \times 10^{-4}$ mol L^{-1} min^{-1}

Q.40 For a linear plot of log (x/m) versus log p in a Freundlich adsorption isotherm, which of the following statements is correct ? (k and n are constants)

(1) Both k and 1/n appear in the slope term (2) 1/n appears as the intercept

(3) Only 1/n appears as the slope (4) log (1/n) appears as the intercept

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Surface chemistry, Exercise. # 1, Q.45*]

[*JEE Advance, Chapter : Surface chemistry, Exercise. # 1, Q.8*]

Ans. **[3]**

Sol. $\frac{x}{m} = KP^{1/n}$

$\log \frac{x}{m} = \frac{1}{n} \log P + \log K$

$\therefore$ Slope is $\frac{1}{n}$

Q.41 Which of the following atoms has the highest first ionization energy ?

(1) Rb (2) Na (3) K (4) Sc

Ans. **[4]**

Sol. Order of first ionization energy : Sc > Na > K > Rb

Q.42 Which one of the following ores is best concentrated by froth floatation method ?

(1) Magnetite (2) Siderite (3) Galena (4) Malachite

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Metallurgy, Exercise. # 4, Q.3*]

[*JEE Advance, Chapter : Metallurgy, Exercise. # 2, Q.43*]

Ans. **[3]**

Sol. Froth floatation method is mainly used for sulphide ores

Magnetite → Fe_3O_4

Siderite → $FeCO_3$

Galena → PbS

Malachite → $Cu(OH)_2 \cdot CuCO_3$

Q.43 Which one of the following statements about water in **FALSE** ?

(1) Water is oxidized to oxygen during photosynthesis

(2) Water can act both as an acid and as a base

(3) There is extensive intramolecular hydrogen bonding in the condensed phase

(4) Ice formed by heavy water sinks in normal water

Ans. **[3]**

Sol. In water, there is intermolecular hydrogen bonding, not intramolecular hydrogen bonding

Q.44 The main oxides formed on combustion of Li, Na and K in excess of air are, respectively :

(1) Li_2O, Na_2O and KO_2 (2) LiO_2, Na_2O_2 and K_2O

(3) Li_2O_2, Na_2O_2 and KO_2 (4) Li_2O, Na_2O_2 and KO_2

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : S-block elements, Exercise. # 1, Q.19]

[JEE Advance, Chapter : S-block elements, Exercise. # 5, Q.2]

Ans. **[4]**

Sol. $2\,Li + \frac{1}{2}O_{2(excess)} \xrightarrow{\Delta} Li_2O$

$2Na + O_{2(excess)} \xrightarrow{\Delta} Na_2O_2$

$K + O_{2(excess)} \xrightarrow{\Delta} KO_2$

Q.45 The reaction of zinc with dilute and concentrated nitric acid, respectively, produces :

(1) N_2O and NO_2 (2) NO_2 and NO (3) NO and N_2O (4) NO_2 and N_2O

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : p-block elements, Exercise. # 5A, Q.29]

[JEE Advance, Chapter : p-block elements, Key concept, Topic :- Properties of nitric acid]

Ans. **[1]**

Sol. $Zn + HNO_{3(dil)} \longrightarrow Zn(NO_3)_2 + NH_4NO_3 + H_2O$

$\downarrow \Delta$

$N_2O\uparrow$

$Zn + HNO_{3(conc.)} \longrightarrow Zn(NO_3)_2 + NO_2\uparrow + H_2O$

Q.46 The pair in which phosphorous atoms have a formal oxidation state of +3 is :

(1) Orthophosphorous and pyrophosphorous acids (2) Pyrophosphorous and hypophosphoric acids

(3) Orthophosphorous and hypophosphoric acids (4) Pyrophosphorous and pyrophosphoric acids

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : p-block elements, Key concept, Topic :- Oxyacids of phosphorus]

Ans. **[1]**

Sol. Orthophosphorous acid is H_3PO_3 and Pyrophosphorous acid is $H_4P_2O_5$. In both compounds, oxidation state of 'P' is +3

Q.47 Which of the following compounds is metallic and ferromagnetic ?

(1) TiO_2 (2) CrO_2 (3) VO_2 (4) MnO_2

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Solid state, Key concept, Topic :- Magnetic properties of crystals]

[JEE Advance, Chapter : Solid state, Key concept, Topic :- Magnetic properties of crystals]

Ans. **[2]**

Sol. Ferromagnetic materials are Fe, Co, Ni, Gd, CrO_2 etc.

Q.48 The pair having the same magnetic moment is :

[At. No, : Cr = 24, Mn = 25, Fe = 26, Co = 27]

(1) $[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$ (2) $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

(3) $[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{2+}$ (4) $[CoCl_4]^{2-}$ and $[Fe(H_2O)_6]^{2+}$

Students may find similar question in CP exercise sheet :

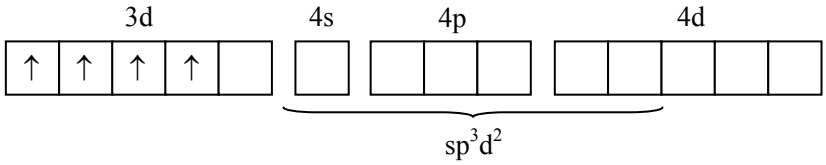
[*JEE Main, Chapter : Co-ordination compounds, Solved Example No.-17*]

[*JEE Advance, Chapter : Co-ordination compounds, Solved Example No.-22*]

Ans. **[2]**

Sol. $[Cr(H_2O)_6]^{+2}$

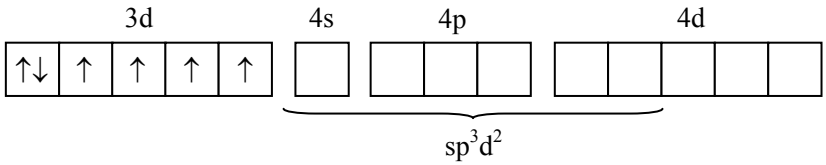
$Cr^{+2} = [Ar]\ 4s^0 3d^4$



It has 4 unpaired electrons

$[Fe(H_2O)_6]^{+2}$

$Fe^{+2} = [Ar]\ 4s^0 3d^6$



It has 4 unpaired electrons

Q.49 Which one of the following complexes shows optical isomerism ?

(1) $[Co(NH_3)_3Cl_3]$ (2) cis$[Co(en)_2Cl_2]Cl$ (3) trans$[Co(en)_2Cl_2]Cl$ (4) $[Co(NH_3)_4Cl_2]Cl$

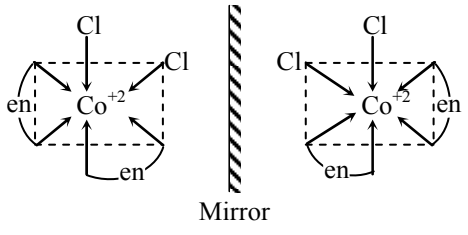
Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Co-ordination compounds, Exercise. # 4, Q.31*]

[*JEE Advance, Chapter : Co-ordination compounds, Exercise. # 3, Q.26*]

Ans. **[2]**

Sol. cis-$[Co(en)_2Cl_2]Cl$ shows optical isomerism. It's general case is $M(aa)_2b_2$



Mirror image is non-super imposable. Hence, this compound is optically active.

Q.50 The concentration of fluoride, lead, nitrate and iron in a water sample from an underground lake was found to be 1000 ppb, 40 ppb, 100 ppm and 0.2 ppm, respectively. This water is unsuitable for drinking due to high concentration of :

(1) Fluoride (2) Lead (3) Nitrate (4) Iron

Ans. **[3]**

Sol.

Substance	Upper limit
Fluoride	2 ppm or 2000 ppb
Lead	0.05 ppm or 50 ppb
Nitrate	50 ppm
Fe	0.2 ppm

In water sample, concentration of nitrate is higher than upper limit. Hence, this water is unsuitable for drinking.

Q.51 The distillation technique most suited for separating glycerol from spent-lye in the soap industry is :

(1) Simple distillation (2) Fractional distillation

(3) Steam distillation (4) Distillation under reduced pressure

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Purification of organic compound, Key concept, Topic :- Distillation under reduced pressure]

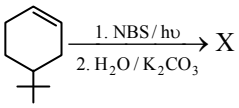
[JEE Advance, Chapter : Practical organic chemistry, Exercise. # 1, Q.12]

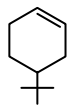
Ans. **[4]**

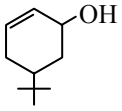
Sol. Distillation under reduced pressure is the technique most suited for separation of Glycerol from Spent-lye (Soap Solution obtain from basic hydrolysis of fats)

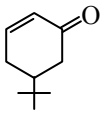
(Mentioned in NCERRT in chapter purification methods)

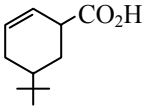
Q.52 The product of the reaction given below is :



(1) 

(2) 

(3) 

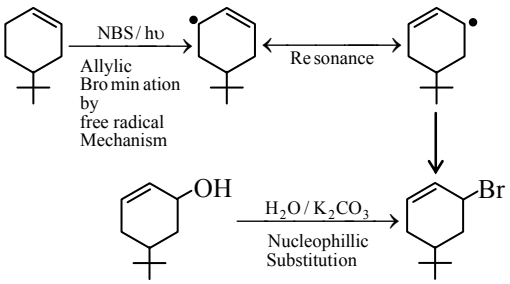
(4) 

Students may find similar question in CP exercise sheet :

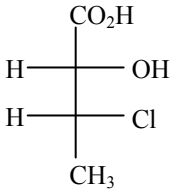
[JEE Advance, Chapter : Hydrocarbon , Exercise. # 3, Q.51]

Ans. **[2]**

Sol.



Q.53 The absolute configuration of



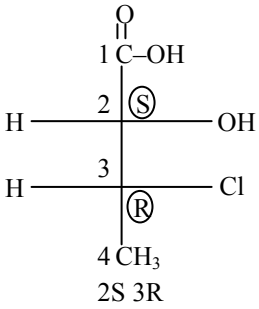
is :

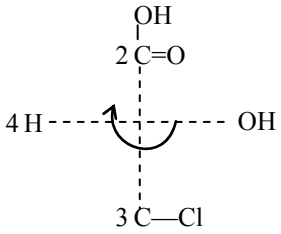
(1) (2R, 3S) (2) (2S, 3R) (3) (2S, 3S) (4) (2R, 3R)

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter : Isomerism, Exercise. # 3, Q.15]
[JEE Advance, Chapter : Isomerism, Exercise. # 1, Q.36]

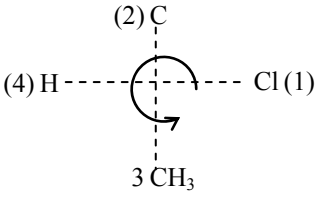
Ans. **[2]**

Sol.





Clockwise but lowest priority group at vertical line so configuration is (S)



Anticlockwise but lowest priority group at vertical line so configuration is (R)

Q.54 2-chloro-2-methylpentane on reaction with sodium methoxide in methanol yields :

(a) $C_2H_5CH_2\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OCH_3$ (b) $C_2H_5CH_2\underset{\displaystyle CH_3}{\overset{}{\underset{|}{C}}}=CH_2$ (c) $C_2H_5CH=\underset{\displaystyle CH_3}{\overset{}{\underset{|}{C}}}-CH_3$

(1) All of these (2) (a) and (c) (3) (c) only (4) (a) and (b)

Students may find same question in CP exercise sheet :
[JEE Main, Chapter : Alkyl halide, Exercise. # 1, Q.23]
[JEE Advance, Chapter : Halo alkane, Exercise. # 1, Q.23]

Ans. **[1]**

Sol.

$$\underset{5}{CH_3}-\underset{4}{CH_2}-\underset{3}{CH_2}-\underset{\displaystyle 1\,CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{\overset{2}{C}}}}}-Cl \xrightarrow[CH_3OH]{NaOCH_3} CH_3-CH_2-CH=\underset{\displaystyle CH_3}{\overset{}{\underset{|}{C}}}-CH_3$$

Elimination

2 Chloro 2 methyl Pentane

$$+\quad CH_3-CH_2-CH_2-\underset{\displaystyle CH_3}{\overset{}{\underset{|}{C}}}=CH_2$$

Elimination

$$+\quad CH_3-CH_2-CH_2-\underset{\displaystyle CH_3}{\overset{}{\underset{|}{C}}}-OCH_3$$

Substitution

Q.55 The reaction of propene with HOCl ($Cl_2 + H_2O$) proceeds through the intermediate :

(1) $CH_3–CH^+–CH_2–OH$ (2) $CH_3–CH^+–CH_2–Cl$ (3) $CH_3–CH(OH)–CH_2^+$ (4) $CH_3–CHCl–CH_2^+$

Students may find same question in CP exercise sheet :

[*JEE Main, Chapter : Hydrocarbon, Exercise. # 1, Q.54*]

[*JEE Advance, Chapter : Hydrocarbon, Exercise. # 1, Q.45*]

Ans. **[2]**

Sol. $CH_3–CH=CH_2 \xrightarrow[(HOCl)]{Cl_2+H_2O} CH_3–\overset{\oplus}{C}H–CH_2Cl \longrightarrow CH_3–CH(\overset{\oplus}{Cl})–CH_2 \xrightarrow{OH^\ominus} CH_3–CH(OH)–CH_2–Cl$

Propene

Ans. is (2) $CH_3–CH^+–CH_2–Cl$

Q.56 In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br_2 used per mole of amine produced are :

(1) One mole of NaOH and one mole of Br_2.

(2) Four moles of NaOH and two moles of Br_2

(3) Two moles of NaOH and two moles of Br_2.

(4) Four moles of NaOH and one mole of Br_2.

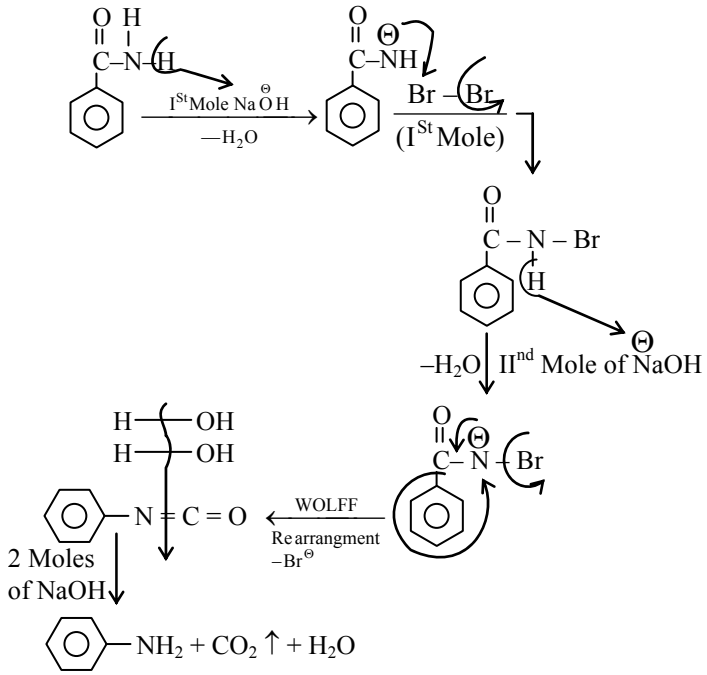
Students may find same question in CP exercise sheet :

[*JEE Main, Chapter : Nitrogen compound, Solved Example- 2*]

[*JEE Advance, Chapter : Nitrogen compound, Exercise. # 4, Q.6*]

Ans. **[4]**

Sol.



Total (4) moles of NaOH and 1 mole of Br_2 is required for conversion from benzamide to per mole of amine

Q.57 Which of the following statements about low density polythene is FALSE?

(1) Its synthesis requires high pressure.

(2) It is a poor conductor of electricity.

(3) Its synthesis requires dioxygen or a peroxide initiator as a catalyst.

(4) It is used in the manufacture of buckets, dust-bins etc.

Students may find similar question in CP exercise sheet :

[*JEE Advance, Chapter : Polymer & Biomolecule, Key concept, Topic:- Addition polymer*]

Ans. **[4]**

Sol. (1) It synthesis requires high pressure : **TRUE**

(2) Low density polythene is not poor conductor of electricity so it is : **TRUE**

(3) It synthesis requires dioxygen or a peroxide initiator as a catalyst : **TRUE**

(4) It is used in the manufacture of buckets, dust-bins etc : **FALSE**

Q.58 Thiol group is present in :

(1) Cytosine (2) Cystine (3) Cysteine (4) Methionine

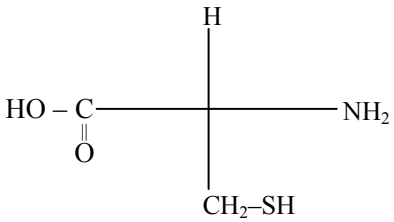
Students may find similar question in CP exercise sheet :

[*JEE Advance, Chapter : Polymer & Biomolecule, Key concept, Topic:- Amino acids*]

Ans. **[3]**

Sol. Thiol (–SH) group is present in cysteine Amino Acid

Structure of Cysteine : -



Q.59 Which of the following is an anionic detergent?

(1) Sodium stearate

(2) Sodium lauryl sulphate

(3) Cetyltrimethyl ammonium bromide

(4) Glyceryl oleate

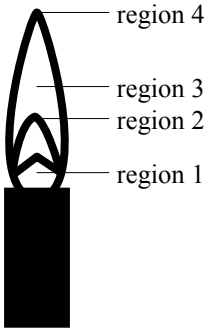
Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter : Polymer & Biomolecule, Key concept, Topic :- 4.9 (i)*]

Ans. **[2]**

Sol. Sodium lauryl salphate is an example of anionic detergent

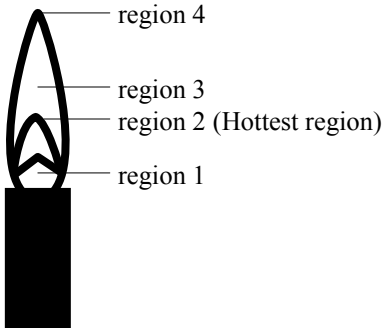
Q.60 The hottest region of Bunsen flame shown in the figure below is :



(1) region 1 (2) region 2 (3) region 3 (4) region 4

Ans. **[2]**

Sol.



The very top of the inner cone is Hottest because when you have a definite inner cone, that's when there is the most air flow through the Bunsen burner. When you have a definite inner cone, that's when there is the most air flow through the Bunsen burner and the most heat and that is concentrated at the tip of inner cone.





Part C – MATHEMATICS

Q.61 If $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$, and $S = \{x \in R : f(x) = f(-x)\}$; then S :

(1) is an empty set (2) contains exactly one element

(3) contains exactly two elements (4) contains more than two elements

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Function, Ex. # 4, Q._17]

Ans. **[3]**

Sol. Given that $f(x) + 2f\left(\frac{1}{x}\right) = 3x, \ x \neq 0$ ....(1)

Replace $x \to \frac{1}{x}$ then $2f(x) + f\left(\frac{1}{x}\right) = \frac{3}{x}$ ...(2)

from (1) & (2)

$f(x) = -x + \frac{2}{x}$

$\therefore \quad f(-x) = x - \frac{2}{x}$

$f(x) = f(-x)$ (Given)

$\Rightarrow \quad -x + \frac{2}{x} = x - \frac{2}{x}$

$\Rightarrow \quad 2x^2 = 4$

$\Rightarrow \quad x^2 = 2$

$\Rightarrow \quad x = \pm\sqrt{2}$

So numbers of solutions is equal to 2.

Q.62 A value of θ for which $\frac{2 + 3i\sin\theta}{1 - 2i\sin\theta}$ is purely imaginary, is :

(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{6}$ (3) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (4) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Complex Number, Exercise # 2, Q.8]

[JEE Advance, Chapter : Complex Number, Exercise # 1, Q.21]

Ans. **[4]**

Sol. For purely imaginary, real part of the given complex number should be zero.

$\therefore \ \text{Re}\left[\frac{2 + 3i\sin\theta}{1 - 2i\sin\theta}\right] = 0$

$$\text{Re}\left[\frac{2+3i\sin\theta}{1-2i\sin\theta}\times\frac{1+2i\sin\theta}{1+2i\sin\theta}\right]=0$$

$$\text{Re}\left[\frac{2-6\sin^2\theta}{1+4\sin^2\theta}+\frac{i7\sin\theta}{1+4\sin^2\theta}\right]=0$$

$$\Rightarrow \frac{2-6\sin^2\theta}{1+4\sin^2\theta}=0$$

$$\Rightarrow \sin^2\theta=\frac{1}{3}$$

$$\Rightarrow \sin\theta=\pm\frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta=\sin^{-1}\left(\frac{1}{\sqrt{3}}\right),\ \sin^{-1}\left(-\frac{1}{\sqrt{3}}\right)$$

Q.63 The sum of all real values of x satisfying the equation $\left(x^2-5x+5\right)^{x^2+4x-60}=1$ is

(1) 3 (2) –4 (3) 6 (4) 5

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Log & Modulus , Exercise # 3, Q.14]

Ans. **[1]**

Sol. Given equation $\left(x^2-5x+5\right)^{x^2+4x-60}=1$

Case I : $(1)^{\text{Any finite real number}}$ = 1

$\therefore (x^2-5x+5)=1$ and $(x^2+4x-60)$ = any finite real number

$\Rightarrow (x^2-5x+4)=0$

$\Rightarrow (x-1)(x-4)=0$

$x=1,4 \Rightarrow x=1,4$ satisfies given equation

Case II : $(-1)^{\text{even integer}}$ = 1

$\therefore (x^2-5x+5)=-1$ and $x^2+4x-60$ = even integer

$\Rightarrow x^2-5x+6=0$

$\Rightarrow x=2,3$

Check (A) : At $x=2 \Rightarrow x^2+4x-60=-48$ = (even integer)

$\Rightarrow x=2$ is root

Check (B) : At $x=3 \Rightarrow x^2+4x-60=-39 \neq$ even integer

$\Rightarrow x=3$ is not root

Case III : (Any non zero real number)^0 = 1

$\therefore (x^2-5x+5)$ is any non zero real number and $x^2+4x-60=0$

$\Rightarrow (x+10)(x-6)=0$

$\Rightarrow x=6,-10$ are the roots

$\therefore$ overall roots are 1, 4, 2, –10, 6

Hence sum of roots = 1 + 4 + 2 + 6 – 10 = 3

Q.64 If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and A adj A = AA^T, then 5a + b is equal to

(1) –1 (2) 5 (3) 4 (4) 13

Students may find similar question in CP exercise sheet :

[*JEE Advance, Chapter : Matrix, Exercise # 5A, Q.28*]

Ans. **[2]**

Sol. $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$

$A.\text{Adj } A = A.A^T$ (given)

$\Rightarrow |A|\,(I_2) = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 5a & 3 \\ -b & 2 \end{bmatrix}$, $\{\text{using property } A.\text{Adj } A = |A| \cdot I_2\}$

$\Rightarrow (10a + 3b) \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 25a^2 + b^2 & 15a - 2b \\ 15a - 2b & 13 \end{bmatrix}$, $\{|A| = 10a + 3b\}$

$\Rightarrow \begin{bmatrix} 10a + 3b & 0 \\ 0 & 10a + 3b \end{bmatrix} = \begin{bmatrix} 25a^2 + b^2 & 15a - 2b \\ 15a - 2b & 13 \end{bmatrix}$

$\Rightarrow 15a - 2b = 0$ ...(1)

$\Rightarrow 10a + 3b = 13$ ...(2)

$\Rightarrow 25a^2 + b^2 = 13$ ...(3)

By solving eq. (1) and (2)

$a = \frac{2}{5}$; $b = 3$

Check eq. (3)

Also At $a = \frac{2}{5}$, $b = 3$

$25a^2 + b^2 = \left(25 \times \frac{4}{25}\right) + 9$

$= 13$

satisfying equation (3)

$\therefore 5a + b = \left(5 \times \frac{2}{5}\right) + 3 = 2 + 3 = 5$

Q.65 The system of linear equations

$x + \lambda y - z = 0$

$\lambda x - y - z = 0$

$x + y - \lambda z = 0$

has a non-trivial solution for

(1) infinitely many values of λ (2) exactly one value of λ

(3) exactly two values of λ (4) exactly three values of λ

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Determinant, Exercise # 1, Q.41]

[JEE Advance, Chapter : Determinant, Exercise # 3, Q.3]

Ans. **[4]**

Sol. For non-trivial solution, $\Delta = 0$

$$\Delta = \begin{vmatrix} 1 & \lambda & -1 \\ \lambda & -1 & -1 \\ 1 & 1 & -\lambda \end{vmatrix} = 0$$

$\Rightarrow 1(\lambda + 1) - \lambda(-\lambda^2 + 1) - 1(\lambda + 1) = 0$

$\Rightarrow \lambda^3 - \lambda = 0$

$\Rightarrow \lambda(\lambda^2 - 1) = 0$

$\Rightarrow \lambda = 0, 1, -1$

$\therefore$ exactly three values of λ exist for which the given system of linear equations has non-trivial solution.

Q.66 If all the words (with or without meaning) having five letters, formed using the letters of the word SMALL and arranged as in a dictionary; then the position of the word SMALL is :

(1) 46th (2) 59th (3) 52nd (4) 58th

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter :P & C, Exercise # 5A, Q.8]

[JEE Advance, Chapter : P & C, Exercise # 5B, Q.3]

Ans. **[4]**

Sol. **Method-I: Short Trick**

S	M	A	L	L

~~A~~ L L M S

S	M	A	L	L

~~A~~ ~~L~~ L M S ↓

$$\frac{(4\times 4!)}{2!} + \frac{(3\times 3!)}{2!} + \frac{(0\times 2!)}{2!} + \frac{(0\times 1!)}{2!} + (1\times 0!)$$

$= 48 + 9 + 0 + 0 + 1$

$= 58^{th}$ rank

Method -II : Actual Method

(1) Words start with A $\Rightarrow$ A _ _ _ _ (S, M, L, L) $= \frac{4!}{2!} = 12$

(2) Words start with L ⇒ L  $= 4! = 24$

(3) Words start with M ⇒ M  $= \frac{4!}{2!} = 12$

(4) Words start with SA ⇒ S A  $= \frac{3!}{2!} = 3$

(5) Words start with SL ⇒ S L  $= 3! = 6$

(6) Words start with SMALL ⇒ | S | M | A | L | L | $= 1$

Total rank = 12 + 24 + 12 +3 + 6 +1

= 58

Q.67 If the number of terms in the expansion of $\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n$, $x \neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion, is

(1) 64 (2) 2187 (3) 243 (4) 729

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Binomial Theorem, Exercise # 1, Q.23]

Ans. **[Bonus]**

Sol. Given expansion

$\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n$, $x \neq 0$,

$\frac{(x^2-2x+4)^n}{x^{2n}}$, $x \neq 0$

No. of terms in given polynomial = 2n + 1 = 28

$n = \frac{27}{2}$ which is not possible

If multinomial theorem is used then

$\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n = \frac{(x^2-2x+4)^n}{x^{2n}}$

then no. of terms = ${}^{n+3-1}C_{3-1} = 28$

$\Rightarrow {}^{n+2}C_2 = 28 \Rightarrow \frac{(n+2)(n+1)}{2!} = 28$

$\Rightarrow (n+2)(n+1) = 8 \times 7$

$\Rightarrow n + 2 = 8$

$\Rightarrow n = 6$

$\therefore$ sum of coefficient (put x = 1) $= \left(1 - \frac{2}{1} + \frac{4}{1}\right)^{n=6} = (3)^6 = 729$

Although, application of multinomial theorem for one variable expression is conceptually wrong, hence bonus must be awarded.

Q.68 If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is

(1) $\frac{8}{5}$ (2) $\frac{4}{3}$ (3) 1 (4) $\frac{7}{4}$

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Progression, Exercise # 1, Q.41]

[JEE Advance, Chapter : Progression, Exercise # 3, Q. 4]

Ans. **[2]**

Sol. Let a be the first term and d is the common difference of A.P.

Now 2nd, 5th and 9th terms of A.P. are

$T_2 = a + d$

$T_5 = a + 4d$

$T_9 = a + 8d$

These terms are in G.P.

$\therefore (a + 4d)^2 = (a + d)(a + 8d)$

$\Rightarrow a^2 + 16d^2 + 8ad = a^2 + 9ad + 8d^2$

$\Rightarrow 8d^2 = ad$

$\Rightarrow 8d = a \qquad (d \neq 0)$

Now common ratio $r = \frac{a + 4d}{a + d} = \frac{8d + 4d}{8d + d} = \frac{4}{3}$

Q.69 If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + ...$, is $\frac{16}{5}$m, then m is equal to

(1) 102 (2) 101 (3) 100 (4) 99

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Progression, Exercise # 5A, Q.24]

Ans. **[2]**

Sol. $\frac{1}{5^2}[8^2 + 12^2 + 16^2 + 20^2 + ... + \text{upto } 10 \text{ terms}] = \frac{16}{5}m$

$\frac{4^2}{5^2}[2^2 + 3^3 + ... + 11^2] = \frac{16}{5}m$

$\frac{1}{5}[(1^2 + 2^2 + ... + 11^2) - 1^2] = m$

$\frac{1}{5}\left[\left(\frac{11\times12\times23}{6} - 1\right)\right] = m$

$(22 \times 23 - 1) = 5m$

$m = 101$

Q.70 Let $p = \lim_{x\to0^+}(1+\tan^2\sqrt{x})^{\frac{1}{2x}}$ then log p is equal to

(1) 2 (2) 1 (3) $\frac{1}{2}$ (4) $\frac{1}{4}$

Ans. **[3]**

Sol. Indeterminate form 1^∞.

So $p = e^{\lim_{x\to0^+}(\tan^2\sqrt{x}).\frac{1}{2x}}$

$\Rightarrow p = e^{1/2}$

$\Rightarrow \log p = \frac{1}{2}$

Q.71 For $x \in R$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then :

(1) g is not differentiable at x = 0

(2) $g'(0) = \cos(\log 2)$

(3) $g'(0) = -\cos(\log 2)$

(4) g is differentiable at x = 0 and $g'(0) = -\sin(\log 2)$

Students may find similar question in CP exercise sheet :

[*JEE Advance, Chapter :AOD, Exercise # 5A, Q.9*]

Ans. **[2]**

Sol. $f(x) = \log 2 - \sin x$, $\{ \because$ when $x \to 0$, $\log 2 - \sin x > 0 \}$

$g(x) = \log 2 - \sin(\log 2 - \sin x)$

$\Rightarrow g'(x) = 0 + \cos(\log 2 - \sin x) \cdot \cos x$

$\Rightarrow g'(0) = \cos(\log 2)$

Q.72 Consider $f(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$, $x \in \left(0, \frac{\pi}{2}\right)$. A normal to $y = f(x)$ at $x = \frac{\pi}{6}$ also passes through the point :

(1) (0, 0) (2) $\left(0, \frac{2\pi}{3}\right)$ (3) $\left(\frac{\pi}{6}, 0\right)$ (4) $\left(\frac{\pi}{4}, 0\right)$

Students may find similar question in CP exercise sheet :
[*JEE Main, Chapter : AOD-1, Exercise # 5A, Q.31*]
[*JEE Advance, Chapter : AOD-1, Exercise # 1, Q. 5*]

Ans. **[2]**

Sol. $f(x) = \tan^{-1}\sqrt{\frac{1+\sin x}{1-\sin x}}, \quad x \in \left(0, \frac{\pi}{2}\right)$

$$f(x) = \tan^{-1}\left|\frac{\cos\frac{x}{2}+\sin\frac{x}{2}}{\cos\frac{x}{2}-\sin\frac{x}{2}}\right|, \qquad \frac{x}{2} \in \left(0, \frac{\pi}{4}\right)$$

$$\Rightarrow f(x) = \tan^{-1}\left(\frac{\cos\frac{x}{2}+\sin\frac{x}{2}}{\cos\frac{x}{2}-\sin\frac{x}{2}}\right)$$

$$\Rightarrow f(x) = \tan^{-1}\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)$$

$$\Rightarrow f(x) = \frac{\pi}{4}+\frac{x}{2}$$

$$\Rightarrow f'(x) = \frac{1}{2}$$

$\Rightarrow$ Slope of tangent $= \frac{1}{2}$

$\Rightarrow$ Slope of normal $= -2$

$\therefore$ Equation of normal at $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ is

$$y - \frac{\pi}{3} = -2\left(x - \frac{\pi}{6}\right)$$

$\Rightarrow y = -2x + \frac{2\pi}{3}$ passes through $\left(0, \frac{2\pi}{3}\right)$

Q.73 A wire of length 2 units is cut into two parts which are bent respectively to form a square of side = x units and a circle of radius = r units. If the sum of the areas of the square and the circle so formed is minimum, then :

(1) $2x = (\pi + 4)r$ (2) $(4 - \pi)x = \pi r$ (3) $x = 2r$ (4) $2x = r$

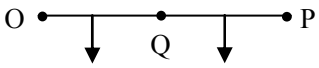
Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : AOD-3 (max.-min.), Exercise # 1, Q.68]

[JEE Advance, Chapter : AOD-3 (max.-min.), Exercise # Board pattern, Q.17]

Ans. **[3]**

Sol.



Circle (Radius = r) units

Sq. (side = x units)

OQ + QP = 2 (given)

$\Rightarrow 4x + 2\pi r = 2$

$\Rightarrow 2x + \pi r = 1$ … (1)

$r = \dfrac{1-2x}{\pi}$

Sum of areas = $A = x^2 + \pi r^2$

$A = x^2 + \pi \dfrac{(1-2x)^2}{\pi^2}$

$\dfrac{dA}{dx} = 2x + \dfrac{2(1-2x)}{\pi}(-2)$

$\Rightarrow 2x - 4r = 0$

$\Rightarrow 4r = 2x$

$\Rightarrow 2r = x$

Q.74 The integral $\int \dfrac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$ is equal to -

(1) $\dfrac{-x^5}{(x^5 + x^3 + 1)^2} + C$ (2) $\dfrac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$

(3) $\dfrac{x^5}{2(x^5 + x^3 + 1)^2} + C$ (4) $\dfrac{-x^{10}}{2(x^5 + x^3 + 1)^2} + C$

where C is an arbitrary constant.

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter :Ind. Integral, Exercise # 4, Q.19]

Ans. **[2]**

Sol. $I = \int \dfrac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$

$$I = \int \frac{\frac{2}{x^3}+\frac{5}{x^6}}{\left(1+\frac{1}{x^2}+\frac{1}{x^5}\right)^3}\,dx$$

$$\left\{\begin{array}{ll} \text{Let} & 1+\frac{1}{x^2}+\frac{1}{x^5}=t \\ & \Rightarrow \left(\frac{-2}{x^3}-\frac{5}{x^6}\right)dx = dt \end{array}\right\}$$

$$\therefore I = \int \frac{-dt}{t^3}$$

$$\Rightarrow I = -\int t^{-3}\,dt$$

$$\Rightarrow I = \frac{-t^{-2}}{-2} + C = \frac{1}{2t^2} + C$$

$$\Rightarrow I = \frac{1}{2\left(1+\frac{1}{x^2}+\frac{1}{x^5}\right)^2} + C$$

$$\Rightarrow I = \frac{x^{10}}{2(x^5+x^3+1)^2} + C$$

Q.75 $\lim\limits_{n\to\infty}\left(\frac{(n+1)(n+2)....3n}{n^{2n}}\right)^{1/n}$ is equal to -

(1) $\frac{18}{e^4}$ (2) $\frac{27}{e^2}$ (3) $\frac{9}{e^2}$ (4) 3 log3 – 2

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter :Def. Integral , Exercise # 5B, Q.27]
[JEE Advance, Chapter : Def. Integral, Exercise # 4,Q.26]

Ans. **[2]**

Sol. $y = \left[\left(\frac{n+1}{n}\right)\left(\frac{n+2}{n}\right).....\left(\frac{n+2n}{n}\right)\right]^{1/n}$

Taking log both sides

$$\log y = \frac{1}{n}\sum_{r=1}^{2n}\log\left(1+\frac{r}{n}\right)$$

$$\Rightarrow \log y = \int_0^2 \log(1+x)\,dx \qquad \left\{\begin{array}{l}\text{Put } 1+x=t \\ \therefore\ dx = dt\end{array}\right\}$$

$\Rightarrow \log y = \int_1^3 \log t\,dt = \left| t\log t - t \right|_1^3$

$\Rightarrow \log y = 3\log 3 - 3 - [-1]$

$\Rightarrow \log y = \log 27 - 2$

$\Rightarrow y = e^{\log 27 - 2}$

$\Rightarrow y = 27e^{-2}$

Q.76 The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is -

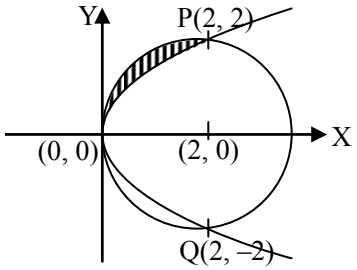
(1) $\pi - \frac{4}{3}$ (2) $\pi - \frac{8}{3}$ (3) $\pi - \frac{4\sqrt{2}}{3}$ (4) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$

Students may find similar question in CP exercise sheet :

[*JEE Advance, Chapter : Area under the curve, Exercise # 5A, Q.2*]

Ans. **[2]**

Sol.



$\Rightarrow y^2 \geq 2x$ ...(1)

$\Rightarrow x^2 + y^2 \leq 4x$

$\Rightarrow x^2 - 4x + y^2 \leq 0$

$\Rightarrow (x-2)^2 + y^2 \leq 2^2$ ...(2)

for point of intersection

$x^2 + 2x = 4x$

$x^2 - 2x = 0$

$x(x-2) = 0$

$x = 0, 2$

$\therefore$ at $x = 0$, $y = 0$

and at $x = 2$, $y = \pm 2$

$\therefore$ P(2, 2) and Q(2, –2)

Hence required area

$$= \int_0^2 (\sqrt{4x - x^2} - \sqrt{2x})\,dx$$

$$= \int_0^2 (\sqrt{4 - (x-2)^2} - \sqrt{2x})\,dx$$

$$= \left[\frac{x-2}{2}\sqrt{4x-x^2} + \frac{4}{2}\sin^{-1}\left(\frac{x-2}{2}\right) - \frac{\sqrt{2}x^{3/2}}{3/2}\right]_0^2$$

$$= \left[-\sqrt{2}\,2^{3/2}\times\frac{2}{3} - \left(\frac{4}{2}\times\frac{-\pi}{2}\right)\right]$$

$$= \left[-\frac{4\times 2}{3} + \pi\right] = \pi - \frac{8}{3}$$

Q.77 If a curve y = f(x) passes through the point (1, –1) and satisfies the differential equation, y(1 + xy)dx = x dy, then $f\left(-\frac{1}{2}\right)$ is equal to -

(1) $-\frac{2}{5}$ (2) $-\frac{4}{5}$ (3) $\frac{2}{5}$ (4) $\frac{4}{5}$

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Differential Equation, Exercise # 3, Page No. 3, Q.13]

Ans. **[4]**

Sol. $y(1 + xy) = x\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{y}{x} + y^2$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = y^2$$

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} - \frac{1}{x}\cdot\frac{1}{y} = 1 \qquad \text{Let } -\frac{1}{y} = t \Rightarrow \frac{1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + \frac{1}{x}.t = 1$$

$$\text{I.F.} = e^{\int\frac{1}{x}dx} = e^{\ell n\, x}$$

I.F. = x

Solution of this differential equation is

$$\Rightarrow t.x = \int x\,dx$$

$$\Rightarrow tx = \frac{x^2}{2} + C$$

$$\Rightarrow -\frac{x}{y} = \frac{x^2}{2} + C \qquad \text{this curve passes through } (1, -1) \text{ so}$$

$$-(-1) = \frac{1}{2} + C \qquad \Rightarrow C = \frac{1}{2}$$

$\Rightarrow -\frac{x}{y} = \frac{x^2}{2} + \frac{1}{2}$ ....(1)

from at $x = -\frac{1}{2}$

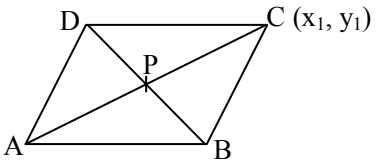
$\frac{1}{2y} = \frac{1}{8} + \frac{1}{2}$ $\Rightarrow \frac{1}{y} = \frac{1}{4} + 1 \Rightarrow y = \frac{4}{5}$

Q.78 Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at $(-1, -2)$, then which one of the following is a vertex of this rhombus ?

(1) $(-3, -9)$ (2) $(-3, -8)$ (3) $\left(\frac{1}{3}, -\frac{8}{3}\right)$ (4) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$

Ans. **[3]**

Sol.



Equation of AB $\Rightarrow x - y + 1 = 0$ … (1)

Equation of AD $\Rightarrow 7x - y - 5 = 0$ … (2)

Solve (1) & (2) $\Rightarrow A(1, 2)$

$\because$ Point $P(-1, -2)$ is the mid-point of the diagonals AC & BD

$\because$ AP = CP $\Rightarrow$ C $\Rightarrow \begin{cases} \frac{x_1 + 1}{2} = -1 \Rightarrow x_1 = -3 \\ \frac{y_1 + 2}{2} = -2 \Rightarrow y_1 = -6 \end{cases}$

C$(-3, -6)$

$\therefore$ Equation of BC $\Rightarrow 7x - y + \lambda_1 = 0$

It's passes through the point C, then

$\Rightarrow 7(-3) - (-6) + \lambda_1 = 0$

$\Rightarrow \lambda_1 = 15$

$\therefore$ Equation of BC is $\Rightarrow 7x - y + 15 = 0$ … (3)

Solve (1) & (3) $\Rightarrow B\left(\frac{-7}{3}, \frac{-4}{3}\right)$

$\therefore$ Equation of CD $\Rightarrow x - y + \lambda_2 = 0$

$\Rightarrow x - y - 3 = 0$ … (4)

Solve (2) & (4) $\Rightarrow D\left(\frac{1}{3}, \frac{-8}{3}\right)$

Q.79 The centres of those circles which touch the circle, $x^2 + y^2 - 8x - 8y - 4 = 0$, externally and also touch the x-axis, lie on :

(1) a circle (2) an ellipse which is not a circle

(3) a hyperbola (4) a parabola

Students may find similar question in CP exercise sheet :

[*JEE Main, Chapter :Circle, Exercise # 4, Q.81*]

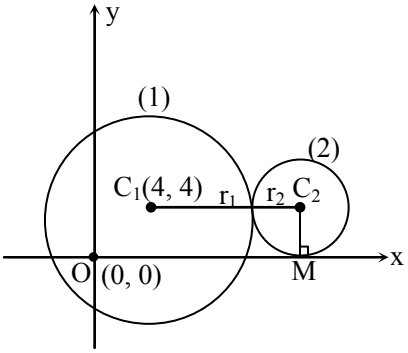
[*JEE Advance, Chapter : Parabola, Exercise # 5B, , Q.7*]

Ans. **[4]**

Sol. $x^2 + y^2 - 8x - 8y - 4 = 0$ … (1)

C_1 = centre (4, 4)

r_1 = radius = $\sqrt{16+16+4} = 6$



Locus of the centre $C_2(h, k)$ = ?

$\because$ radius of circle (2) is

$r_2 = |k|$

& circle (2) touches the circle (1) externally,

So, $C_1C_2 = r_1 + r_2$

$\Rightarrow (h+4)^2 + (k-4)^2 = (6 + |k|)^2$

$\Rightarrow h^2 + 16 - 8h + k^2 + 16 - 8k = 36 + k^2 + 12|k|$

$\Rightarrow$ Replaced h by x & k by y then

$\Rightarrow x^2 - 8x - 8y - 12|y| - 4 = 0$

$$\because |y| \Rightarrow \begin{cases} y; \text{If } y \geq 0 \\ \text{or} \\ -y; \text{If } y < 0 \end{cases}$$

Case (I) : If $y \geq 0$

$\Rightarrow x^2 - 8x - 20y - 4 = 0$

Case (II) : If $y < 0$

$\Rightarrow x^2 - 8x + 4y - 4 = 0$

Both of these equations are the equation of parabola.

Q.80 If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at (–3, 2), then the radius of S is -

(1) $5\sqrt{2}$ (2) $5\sqrt{3}$ (3) 5 (4) 10

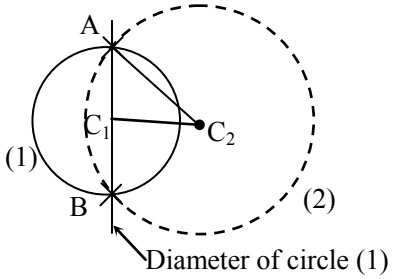
Students may find similar question in CP exercise sheet :

[JEE Main, Chapter :Circle, Exercise # 5A, Q.13]

[JEE Advance, Chapter : Circle, Exercise # 5A, Q.18]

Ans. **[2]**

Sol.



$x^2 + y^2 - 4x + 6y - 12 = 0$ … (1)

centre C_1 (2, –3)

radius $r_1 = 5$

& centre of circle (2) is C_2 (–3, 2)

$\therefore C_1C_2 = \sqrt{(2+3)^2 + (-3-2)^2} = \sqrt{25+25} = 5\sqrt{2}$

$\therefore$ radius (r_2) of circle (2) is

$\Rightarrow r_2 = C_2A = \sqrt{C_1A^2 + C_1C_2^2} = \sqrt{5^2 + (5\sqrt{2})^2}$

$= \sqrt{25+50} = \sqrt{75} = 5\sqrt{3}$

Q.81 Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is -

(1) $x^2 + y^2 - 4x + 8y + 12 = 0$ (2) $x^2 + y^2 - x + 4y - 12 = 0$

(3) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$ (4) $x^2 + y^2 - 4x + 9y + 18 = 0$

Ans. **[1]**

Sol. $y^2 = 8x$

Let, the point P ($2t^2$, 4t) lies on the parabola & centre of the given circle is C (0, –6)

$\therefore$ Distance between the points C & P.

$D = \sqrt{4 + 4 + (4t+6)^2}$

For minimum distance

$\Rightarrow \frac{dD}{dt} = \frac{4(4+3) + 2(4t+6)\times 4}{2\sqrt{4t^4 + 4t}} = 0$

$4t^3 + 8t + 12 = 0$

$t^3 + 2t + 3 = 0$

$t^2(t + 1) – t(t + 1) + 3(t + 1) = 0$

$(t^2 – t + 3)(t + 1) = 0$

$t = –1$

$\therefore$ point P $(2, –4)$

centre C $(0, –6)$

$\therefore$ CP = radius of circle = $\sqrt{4+4} = 2\sqrt{2}$

$\therefore$ Equation of circle $\Rightarrow$ $(x – 2)^2 + (y + 4)^2 = 8$

$x^2 – 4x + 4 + y^2 + 8y + 16 = 8$

$x^2 + y^2 – 4x + 8y + 12 = 10$

Q.82 The eccentricity of the hyperbola whose length of the latus rectum is equal to 8 and the length of its conjugate axis is equal to half of the distance between its foci, is -

(1) $\frac{4}{3}$ (2) $\frac{4}{\sqrt{3}}$ (3) $\frac{2}{\sqrt{3}}$ (4) $\sqrt{3}$

Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : Hyperbola, Exercise # 1, Q.1]

Ans. **[3]**

Sol. Let eqn. of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

It's length of latus-rectum is $\frac{2b^2}{a} = 8$

$\Rightarrow b^2 = 4a$ …(1)

& length of it's conjugate axis = half of the distance between it's foci

$\Rightarrow 2b = \frac{1}{2}(2ae)$

$\Rightarrow b = \frac{ae}{2}$ …(2)

$\because b^2 = a^2(e^2 – 1)$ …(3)

By eq^n (2) & (3)

$\Rightarrow \frac{a^2e^2}{4} = a^2(e^2 – 1)$

$\Rightarrow e^2 - \frac{e^2}{4} = 1$

$\Rightarrow \frac{3e^2}{4} = 1$

$\Rightarrow e = \frac{2}{\sqrt{3}}$

Q.83 The distance of the point (1, –5, 9) from the plane x – y + z = 5 measured along the line x = y = z is :

(1) $3\sqrt{10}$ (2) $10\sqrt{3}$ (3) $\frac{10}{\sqrt{3}}$ (4) $\frac{20}{3}$

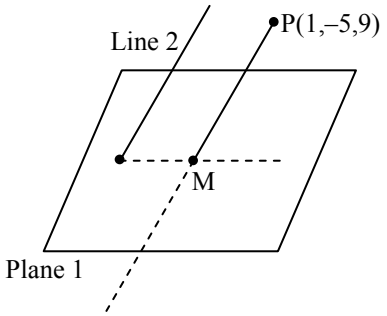
Students may find similar question in CP exercise sheet :

[JEE Main, Chapter :3D, Exercise #3, Q.18]

Ans. **[2]**

Sol. Given plane x – y + z = 5 …(1)

and line x = y = z …(2)



∵ Given line is parallel to line PM

∴ DR's of line PM = DR's of given line

= 1, 1, 1

∴ Eqn. of line PM is $\Rightarrow \frac{x-1}{1}=\frac{y+5}{1}=\frac{z-9}{1}=\lambda$ (Let)

∴ Coordinates of point M is (λ + 1, λ – 5, λ + 9)

∵ Point M lies on the plane x – y + z = 5

That's why $\Rightarrow (\lambda+1)-(\lambda-5)+(\lambda+9)=5$

$\Rightarrow \lambda+15=5$

$\lambda=-10$

∴ M(–9, –15, –1)

∴ $PM=\sqrt{(1+9)^2+(-5+15)^2+(9+1)^2}=\sqrt{100+100+100}$

$=10\sqrt{3}$

Q.84 If the line, $\frac{x-3}{2}=\frac{y+2}{-1}=\frac{z+4}{3}$ lies in the plane, $lx + my – z = 9$, then $l^2 + m^2$ is equal to :

(1) 26 (2) 18 (3) 5 (4) 2

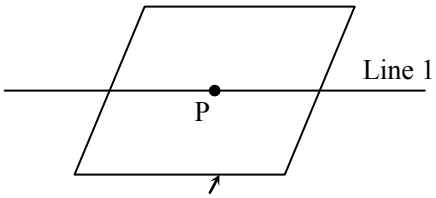
Students may find similar question in CP exercise sheet :

[JEE Main, Chapter :3D, Exercise #4, Q.20]

[JEE Advance, Chapter : 3D, Exercise # 3, Q.21]

Ans. **[4]**

Sol.



$\frac{x-3}{2}=\frac{y+2}{-1}=\frac{z+4}{3}$ ...(1) lie on plane

Let point P lies on line (1)

$\therefore$ P(3, –2, –4)

$\therefore$ Point P also lies on plane $lx + my - z = 9$ ....(2)

$3l - 2m + 4 = 9 \Rightarrow 3l - 2m = 5$ …(3)

$2l - m - 3 = 0 \Rightarrow 4l - 2m = 6$ …(4)

On solving eq. (3) & (4)

$$- \quad + \quad -$$

$$-l = -1$$

$$\Rightarrow l = 1$$

$$m = -1$$

$\therefore l^2 + m^2 = 2$

Q.85 Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a}\times(\vec{b}\times\vec{c})=\frac{\sqrt{3}}{2}(\vec{b}\times\vec{c})$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is :

(1) $\frac{3\pi}{4}$ (2) $\frac{\pi}{2}$ (3) $\frac{2\pi}{3}$ (4) $\frac{5\pi}{6}$

Students may find similar question in CP exercise sheet :
[JEE Main, Chapter : Vector, Exercise # 5A, Q.4]
[JEE Advance, Chapter : Vector, Exercise # 1, Q.29]

Ans. **[4]**

Sol. $\vec{a}\times(\vec{b}\times\vec{c})=\frac{\sqrt{3}}{2}(\vec{b}+\vec{c})$

$(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}=\frac{\sqrt{3}}{2}\vec{b}+\frac{\sqrt{3}}{2}\vec{c}$

Now comparing both sides ($\vec{b}$ and $\vec{c}$ are not parallel)

$\Rightarrow \vec{a}\cdot\vec{c}=\frac{\sqrt{3}}{2}$ & $\vec{a}\cdot\vec{b}=-\frac{\sqrt{3}}{2}$

$\because |\vec{a}| = |\vec{b}| = 1$ and if angle between $\vec{a}$ and $\vec{b}$ is θ then

$$\cos\theta = -\frac{\sqrt{3}}{2}$$

$$\theta = \frac{5\pi}{6}$$

Q.86 If the standard deviation of the numbers 2, 3, a and 11 is 3.5, then which of the following is true ?

(1) $3a^2 - 26a + 55 = 0$ (2) $3a^2 - 32a + 84 = 0$

(3) $3a^2 - 34a + 91 = 0$ (4) $3a^2 - 23a + 44 = 0$

Ans. **[2]**

Sol. Observations are 2, 3, a, 11

Mean $\bar{x} = \frac{2+3+a+11}{4} = 4 + \frac{a}{4}$ and $\sigma^2 = (3.5)^2 = 12.25$

Now, $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - (\bar{x})^2$

$$12.25 = \frac{1}{4}[4+9+a^2+121] - \left(4+\frac{a}{4}\right)^2$$

$$\Rightarrow 12.25 = \frac{134+a^2}{4} - 16 - \frac{a^2}{16} - 2a$$

$$\Rightarrow 12.25 = 33.5 + \frac{a^2}{4} - 16 - \frac{a^2}{16} - 2a$$

$$\Rightarrow \frac{3a^2}{16} - 2a + 5.25 = 0$$

$$\Rightarrow 3a^2 - 32a + 84 = 0$$

Q.87 Let two fair six-faced dice A and B be thrown simultaneously. If E_1 is the event that die A shows up four, E_2 is the event that die B shows up two and E_3 is the event that the sum of numbers on both dice is odd, then which of the following statements is NOT true ?

(1) E_1 and E_2 are independent (2) E_2 and E_3 are independent

(3) E_1 and E_3 are independent (4) E_1, E_2 and E_3 are independent

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Probability, Exercise # 4, Q.27]

Ans. **[4]**

Sol. $P(E_1) = \frac{1}{6}$ E_1 = shows 4

$P(E_2) = \frac{1}{6}$ E_2 = shows 2

$P(E_3) = \frac{18}{36}$ E_3 = sum equal to odd on both dice

$P(E_1 \cap E_2) = \frac{1}{36}$ $E_1 \cap E_2 = \{(4, 2)\}$

$P(E_1 \cap E_3) = \frac{3}{36}$ $E_1 \cap E_3 = \{(4, 1), (4, 3), (4, 5)\}$

$P(E_2 \cap E_3) = \frac{3}{36}$ $E_2 \cap E_3 = \{(2, 1), (2, 3), (2, 5)\}$

$E_1 \cap E_2 \cap E_3 = \{\ \}$

Check by options

(1) $P(E_1 \cap E_2) = P(E_1) \times (E_2)$

$\frac{1}{36} = \frac{1}{6} \times \frac{1}{6}$ $\Rightarrow E_1$ and E_2 are independent $\rightarrow$ True

(2) $P(E_1 \cap E_2) = P(E_1) \times P(E_2)$

$\frac{1}{36} = \frac{1}{6} \times \frac{1}{6}$ $\Rightarrow E_1$ and E_2 are independent $\rightarrow$ True

(3) $P(E_1 \cap E_3) = P(E_1) \times P(E_3)$

$\frac{3}{36} = \frac{1}{6} \times \frac{18}{36}$ $\Rightarrow E_1$ and E_3 are independent $\rightarrow$ True

(4) $P(E_1 \cap E_2 \cap E_3) \neq P(E_1) \times P(E_2) \times P(E_3)$

$0 \neq \frac{1}{6} \times \frac{1}{6} \times \frac{18}{36}$ $\Rightarrow E_1, E_2, E_3$ are not independent $\rightarrow$ False

Q.88 If $0 \le x < 2\pi$, then the number of real values of x, which satisfy the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$, is :

(1) 3 (2) 5 (3) 7 (4) 9

Students may find similar question in CP exercise sheet :

[JEE Advance, Chapter : Trigonometric equation, Exercise # 1,Q.12]

Ans. **[3]**

Sol. $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$

$\Rightarrow \frac{\cos\left[x + \left(\frac{4-1}{2}\right)x\right] \cdot \sin\left(\frac{4x}{2}\right)}{\sin\frac{x}{2}} = 0$ [By using series formula]

$\Rightarrow \cos\frac{5x}{2} \cdot \sin 2x = 0$ But $\left(\sin\frac{x}{2} \neq 0\right)$

$\therefore \sin 2x = 0$ | $\cos\frac{5x}{2} = 0$

$\Rightarrow 2x = n\pi$ | $\Rightarrow \frac{5x}{2} = (2n+1)\frac{\pi}{2}$

$\Rightarrow x = \frac{n\pi}{2},\ n \in I$ | $\Rightarrow x = (2n+1)\frac{\pi}{5},\ n \in I$

$\Rightarrow x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ | $x = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$

But $x = 0, 2\pi$ does not satisfy ($\because \sin\frac{x}{2} \neq 0$)

$\therefore$ Solutions are $x = \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \frac{\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{9\pi}{5}$

7 solutions

Q.89 A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30°. After walking for 10 minutes from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60°. Then the time taken (in minutes) by him, from B to reach the pillar, is :

(1) 6 (2) 10 (3) 20 (4) 5

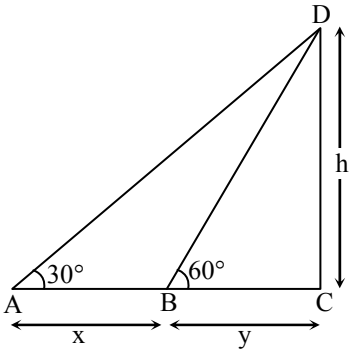
Students may find similar question in CP exercise sheet :

[JEE Main, Chapter : _Height and distance, Exercise # 1,Q.21]

[JEE Advance, Chapter : Height and distance, Exercise # 4,Q.7]

Ans. **[4]**

Sol.



In ΔACD, $\tan 30° = \frac{h}{x+y}$

$\Rightarrow h = \frac{x+y}{\sqrt{3}}$

Now in ΔBCD, $\tan 60° = \frac{h}{y}$

$\Rightarrow \sqrt{3}y = h \Rightarrow \sqrt{3}y = \frac{x+y}{\sqrt{3}} \Rightarrow x = 2y \Rightarrow y = \frac{x}{2}$

Given that travelling time x = 10 min

$\therefore$ travelling time $y = \frac{10}{2} = 5$ min

Q.90 The Boolean Expression $(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$ is equivalent to :

(1) $\sim p \wedge q$ (2) $p \wedge q$ (3) $p \vee q$ (4) $p \vee \sim q$

Ans. **[3]**

Sol.

p	q	~p	~q	$(p \wedge \sim q)$	(1) $(\sim p \wedge q)$	$(p \wedge \sim q) \vee q$	$(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$	(2) $p \wedge q$	(3) $p \vee q$	(4) $(p \vee \sim q)$
T	T	F	F	F	F	T	T	T	T	T
T	F	F	T	T	F	T	T	F	T	T
F	T	T	F	F	T	T	T	F	T	T
F	F	T	T	F	F	F	F	F	F	T

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